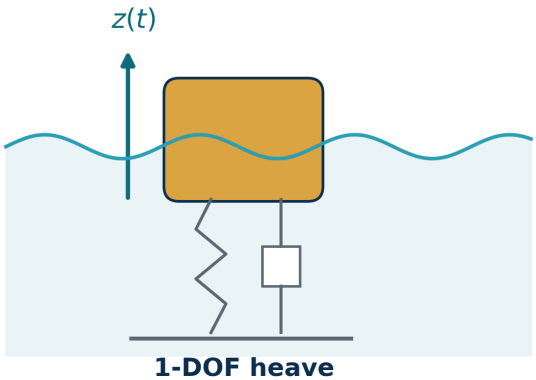
HMEC MATLAB RAO Teaching Demos

Detailed User and Instructor Manual

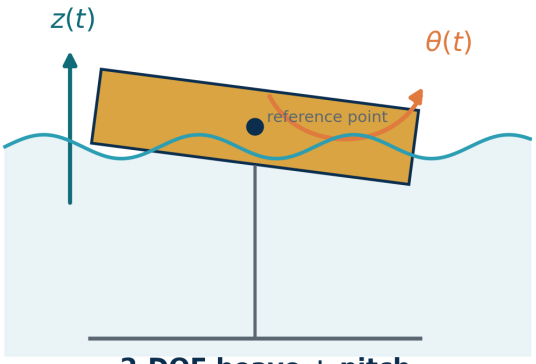
1-DOF Heave and Coupled 2-DOF Heave-Pitch

Equivalent physical models represented by the two demos



1-DOF heave

Spring/dashpot are equivalent restoring and damping symbols.



2-DOF heave + pitch

Translation and rotation are solved through a coupled generalized impedance.

Author	Prof. Ersegun Deniz Gedikli
Affiliation	Department of Ocean and Resources Engineering, University of Hawai‘i at Mānoa
Program	Hawai‘i Marine Energy Center (HMEC)
Manual version	1.0
Date	August 2026
Software files	`HMEC_heave_RAO_demo.m`, `HMEC_heave_pitch_RAO_demo.m`, and `HMEC_RAO_demo_launcher.m`

How to Cite These Demos

When these MATLAB demos, their figures, or modified versions are used in teaching, reports, workshops, or publications, the following APA-style software reference is recommended.

Recommended APA 7 software reference

Gedikli, E. D. (2026). *HMEC MATLAB RAO teaching demos: 1-DOF heave and 2-DOF heave-pitch* (Version 1.0) [Computer software]. Hawai‘i Marine Energy Center, University of Hawai‘i at Mānoa.
Parenthetical in-text citation: (Gedikli, 2026)
Narrative in-text citation: Gedikli (2026)

The manual itself may be cited separately when its explanations, procedures, tables, or verification material are used.

Recommended APA 7 manual reference

Gedikli, E. D. (2026). *User manual for the HMEC MATLAB RAO teaching demos: 1-DOF heave and 2-DOF heave-pitch* (Version 1.0). Hawai‘i Marine Energy Center, University of Hawai‘i at Mānoa.

BibTeX-ready entries

```
@software{gedikli_hmec_rao_demos_2026,
  author       = {Gedikli, Ersegun Deniz},
  title        = {HMEC MATLAB RAO Teaching Demos:
                  1-DOF Heave and 2-DOF Heave-Pitch},
  year         = {2026},
  version      = {1.0},
  organization = {Hawaii Marine Energy Center,
                  University of Hawaii at Manoa},
  note         = {Computer software}
}

@manual{gedikli_hmec_rao_manual_2026,
  author       = {Gedikli, Ersegun Deniz},
  title        = {User Manual for the HMEC MATLAB RAO Teaching Demos:
                  1-DOF Heave and 2-DOF Heave-Pitch},
  year         = {2026},
  edition      = {Version 1.0},
  organization = {Hawaii Marine Energy Center,
                  University of Hawaii at Manoa}
}
```

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Document Control and Scope

Table 1: Document control information.

Item	Description
Document title	Detailed User and Instructor Manual for the HMEC MATLAB RAO Teaching Demos
Version	1.0
Release date	August 2026
Primary audience	Instructors, students, short-course participants, and researchers learning linear wave-body response
Companion lecture	*Marine Hydrodynamics & Floating Body Dynamics*, HMEC Short Course
Demos covered	1-DOF heave RAO; coupled 2-DOF heave-pitch RAO
Units	SI units internally; angular motion calculated in radians
Hydrodynamic inputs	Smooth, physically constrained teaching curves; not device-validated BEM data

Important modeling notice

The demonstrations are educational frequency-domain models. Their hydrodynamic coefficients are synthetic teaching curves with realistic scale and internally consistent matrix properties. They are not predictions for a particular device, are not a substitute for a converged boundary-element model or experiment, and must not be used as the sole basis for engineering design, certification, structural loading, or power-performance claims.

What the animation represents

The spring and dashpot shown in the heave animation are symbols for equivalent linear restoring and damping. They are not necessarily literal components mounted beneath a real floating body. The time histories and animation reconstruct the steady harmonic response at one selected wave period; they are not transient integrations from initial conditions.

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Chapter 1

Purpose, Learning Goals, and Package Organization

1.1 Purpose of the demonstrations

The package turns the frequency-domain equations introduced in the HMEC marine hydrodynamics lecture into two interactive classroom experiments. The first demo isolates heave so students can connect mass, added mass, damping, restoring, excitation, phase, resonance, and response amplitude operator (RAO). The second replaces the scalar denominator by a coupled 2×2 complex impedance matrix and demonstrates how heave and pitch exchange force and moment through off-diagonal terms.

The intended teaching sequence is:

1. derive or review the equation of motion;
2. identify where each coefficient comes from;
3. ask students to predict a trend;
4. change one parameter in the interface;
5. compare RAO magnitude, phase, coefficient curves, motion, and PTO power; and
6. return to the equation to explain the observation.

1.2 Learning objectives

After using both demos, a student should be able to:

- distinguish body mass from hydrodynamic added mass;
- explain why radiation damping exists even in inviscid potential-flow theory;
- compute hydrostatic heave stiffness from waterplane area;
- assemble and interpret scalar and matrix complex impedances;
- report both RAO magnitude and phase;
- predict how mass, inertia, damping, and stiffness shift or reshape response peaks;
- explain why wave amplitude changes physical motion and ideal linear PTO power but not a linear RAO;
- identify when pitch can occur without a directly applied excitation moment;
- state the units of translation-rotation coupling coefficients; and
- recognize the limits of synthetic coefficients and the additional checks required for BEM data.

1.3 Relationship to the companion presentation

The 1-DOF application is designed to follow the presentation section that asks students to build a heave RAO from hydrostatic stiffness, effective mass, damping, dynamic stiffness, RAO magnitude and phase, and one regular-wave response. The coupled demo follows the heave-pitch matrix equation and its frequency-domain solution. The demos therefore support the lecture flow rather than replacing the derivation.

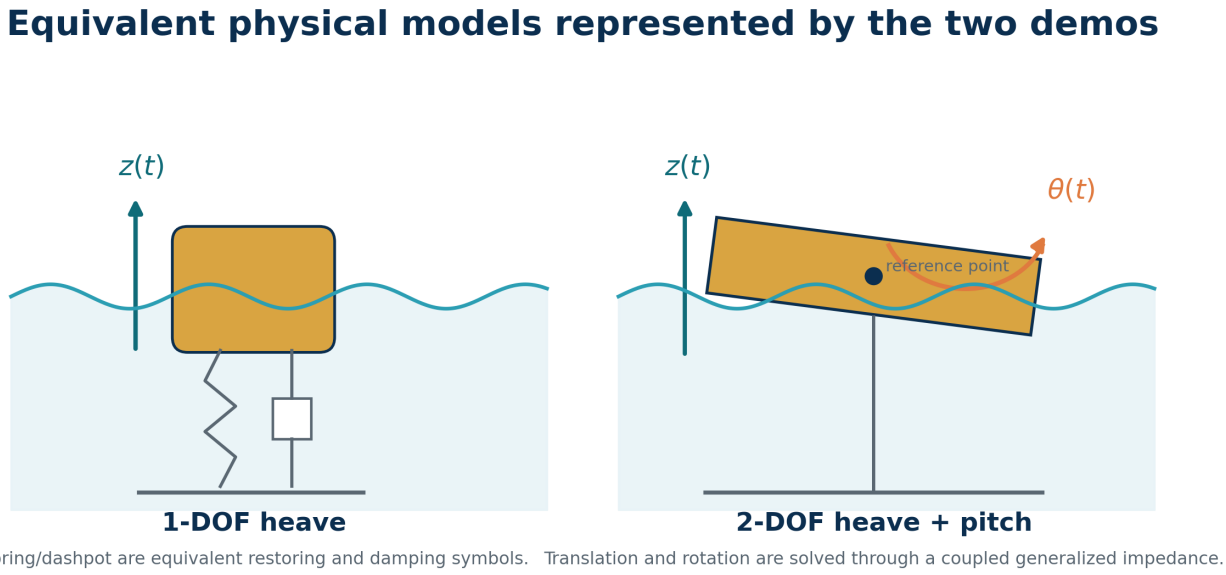


Figure 1.1: Equivalent physical models represented by the two demos. The spring and dashpot are equivalent linear elements. The coupled model translates in heave and rotates in pitch about a selected reference point.

1.4 Package contents

Table 1.1: Core files supplied with the demo package.

File	Purpose
`HMEC_RAO_demo_launcher.m`	Opens a compact launcher with buttons for both interfaces and a combined numerical self-test.
`HMEC_heave_RAO_demo.m`	Runs the interactive 1-DOF heave RAO demonstration. Calling it with `'selftest'` runs numerical checks without creating the GUI.
`HMEC_heave_pitch_RAO_demo.m`	Runs the interactive coupled heave-pitch demonstration. Calling it with `'selftest'` runs matrix and response checks without creating the GUI.
`HMEC_numerical_validation.txt`	Records independent numerical checks of the formulas and expected parameter trends.
`README_HMEC_RAO_Demos.md`	Concise quick-start notes, default values, and implementation cautions.
`HMEC_presentation_recommendations.md`	Suggested scientific and notation refinements for the companion lecture.
This manual	Detailed installation, theory, interface operation, teaching procedures, validation, BEM-data replacement, customization, and troubleshooting.

1.5 Coordinate, sign, and unit conventions

Table 1.2: Conventions used throughout the demos.

Quantity	Convention	Meaning
Heave z	Positive upward	Translation mode 3.
Pitch θ	Right-hand rotation about the transverse y-axis	Rotation mode 5; internally in radians.
Wave elevation η	Positive upward	Incident regular-wave amplitude is $a = \lvert\hat{a}\rvert$.
Harmonic convention	$\Re\{\hat{x}e^{\mathrm{i}\omega t}\}$	Velocity is $\mathrm{i}\omega\hat{x}$; acceleration is $-\omega^2\hat{x}$.
Force vector	$[F_3\ M_5]^{\mathsf{T}}$	Translation component is a force; rotation component is a moment.
Units	SI internally	Input labels display tonne, kN, or MN where convenient; code converts to kg, N, and N m.

1.6 What the package is and is not

Appropriate uses

The demos are well suited for classroom prediction questions, sensitivity studies, coefficient interpretation, introduction to RAOs, explanation of phase, and preparation for importing BEM results from tools such as Capytaine, WAMIT, Nemoh, or AQWA.

Uses that require another model

The demos do not model irregular-wave spectra, radiation memory in a true time-domain simulation, nonlinear drag, slamming, breaking waves, end stops, saturation, nonlinear moorings, generator efficiency, control constraints, structural stress, or fatigue. Those questions require an extended time-domain, CFD, experimental, or coupled engineering model.

Chapter 2

Installation, Quick Start, and Pre-Class Checks

2.1 System requirements

The files use MATLAB user-interface and plotting functions such as `uifigure`, `uigridlayout`, `uispinner`, `uitabgroup`, and `uiaxes`. The code itself does not call a specialized toolbox. A recent MATLAB release is recommended because app-layout behavior and graphics support have improved over time.

The runtime computer should have:

- MATLAB with desktop graphics support;
- permission to execute local `.m` files;
- enough screen resolution to display a large two-column interface; and
- all three core `.m` files in the same folder or on the MATLAB path.

2.2 Installation

1. Create a folder, for example `HMEC_RAO_Demos`.
2. Copy the following files into it:
   - `HMEC_RAO_demo_launcher.m`
   - `HMEC_heave_RAO_demo.m`
   - `HMEC_heave_pitch_RAO_demo.m`
3. In MATLAB, make that folder the **Current Folder**, or add it to the MATLAB path.
4. Run the launcher command shown in Listing 2.1.

```
HMEC_RAO_demo_launcher
```

Listing 2.1: Starting the launcher.

2.3 Opening each demo directly

```
HMEC_heave_RAO_demo
HMEC_heave_pitch_RAO_demo
```

Listing 2.2: Direct commands for the two demos.

Running the direct command is useful when the lecture has already introduced one model and the launcher would add an unnecessary step.

2.4 Run the self-tests before class

The self-tests execute before any user-interface objects are created. They therefore separate the numerical model from environment-specific GUI problems.

```
report1 = HMEC_heave_RAO_demo('selftest')
report2 = HMEC_heave_pitch_RAO_demo('selftest')
```

Listing 2.3: Recommended numerical checks before class.

Expected behavior:

- MATLAB prints a report structure or a set of values without an error;
- the heave test confirms that added mass and radiation damping are finite and nonnegative, the restoring stiffness is positive, increased mass shifts the response toward longer period, increased stiffness shifts it toward shorter period, increased damping lowers the peak, and wave amplitude does not change the RAO;
- the coupled test confirms symmetry, positive-semidefinite hydrodynamic matrices, positive restoring stiffness, correct coupling transfer, expected inertia/stiffness/damping trends, and wave-amplitude invariance.

Honest verification status

The equations and parameter trends were independently reproduced outside MATLAB and passed the checks summarized in Chapter A. A MATLAB or Octave GUI runtime was not available in the manual-generation environment. The embedded self-tests and a launch on the actual classroom computer are therefore the final release-specific verification steps.

2.5 Launcher overview

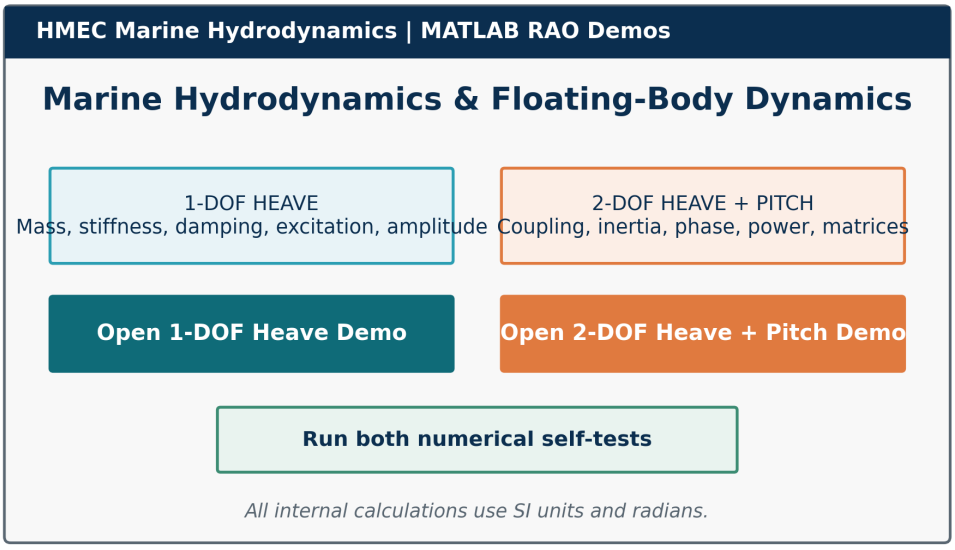


Figure 2.1: Schematic of the launcher. The actual MATLAB window uses the same three actions: open the heave demo, open the coupled demo, or run both self-tests.

2.6 Recommended pre-class checklist

1. Run both commands in Listing 2.3.
2. Open both interfaces and confirm that all tabs render.
3. Move one spinner in each demo and verify that the plots update.
4. Click **Set baseline**; change one parameter; verify that a dashed comparison curve appears.
5. Click **Play motion**; verify that the animation completes.
6. Confirm that the computer is connected to the projector at the same display scaling that will be used in class.
7. Reset both demos to their default values.
8. Keep a static PDF of the default plots available as a backup.

2.7 What to do after an error

Start with the smallest diagnostic:

1. run the relevant `'selftest'` command;
2. use `which HMEC_heave_RAO_demo -all` to confirm MATLAB is calling the intended file;
3. remove older copies from the MATLAB path;
4. restart MATLAB if an old function definition remains cached; and
5. consult Chapter 9 for symptom-specific guidance.

Recommended Classroom Workflow

3.1 Use prediction before interaction

The interface becomes most useful when students make a directional prediction before the instructor changes a value. Otherwise, it can become a passive plotting tool.

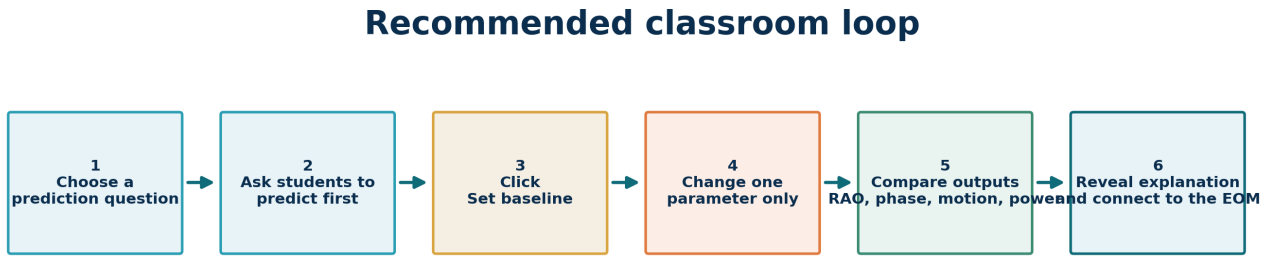


Figure 3.1: Recommended predict-change-interpret classroom loop.

3.2 The baseline-comparison feature

Set baseline stores the current response so it can be shown as a dashed comparison curve. The current response remains solid. A disciplined sequence is:

1. reset the demo;
2. set the baseline;
3. change one parameter only;
4. ask which part of the equation changed;
5. compare peak location, peak magnitude, bandwidth, phase, and power; and
6. return to the original baseline before starting the next question.

Changing several controls at once is useful for open exploration, but it is a poor first method for teaching cause and effect.

3.3 The selected wave period

The selected period serves three roles:

- a vertical marker on the frequency-response plots;
- the frequency used to interpolate hydrodynamic coefficients and complex response; and
- the frequency used to reconstruct the motion animation and time histories.

Move the selected-period control after the class has examined the full RAO. This lets students compare a response below resonance, near resonance, and above resonance without changing the physical model.

3.4 Suggested short teaching scripts

3.4.1 Ten-minute heave script

1. Open the heave demo and identify the four groups in the EOM: inertia, damping, restoring, and excitation.
2. Ask what happens when body mass increases.
3. Set the baseline, increase mass by approximately 40%, and compare peak period.
4. Reset, set the baseline, and increase PTO damping.
5. Compare RAO peak and mean mechanical PTO power; emphasize that maximum motion and maximum power are not identical objectives.
6. Double wave amplitude and show that the RAO does not change although motion and ideal damper power do.

3.4.2 Fifteen-minute heave-pitch script

1. Open the coupled demo and identify z, θ, force, and moment.
2. Set coupling to 0% and explain why the matrix becomes block diagonal.
3. With coupling at 0%, turn off direct pitch excitation and confirm zero pitch.
4. Restore coupling while keeping direct pitch excitation off; observe pitch generated by transfer through the off-diagonal impedance.
5. Increase I_{yy}, then pitch stiffness, and compare opposite period shifts.
6. Switch pitch units from deg/m to rad/m and emphasize that only the display scale changes.

3.5 Questions to ask while plots are visible

Prompt set for both demos

- Did the peak move, change height, change width, or all three?
- Did the phase transition move with the magnitude peak?
- Which coefficient or matrix entry changed directly?
- Which output changed indirectly through coupling?
- Is the visible response peak exactly the same as the simple inertia-restoring balance? Why might it differ?
- Did the control change the RAO, the physical response amplitude, or both?

- What additional physics could reverse or limit this trend in a real device?

3.6 Common misconceptions to address immediately

Table 3.1: Misconceptions the demos are designed to expose.

Misconception	Correction
Added mass is extra ballast attached to the body.	It is a hydrodynamic reaction coefficient representing fluid accelerated by the body motion.
Radiation damping is viscous drag.	It is energy carried away by radiated waves and exists in inviscid potential-flow theory.
A larger RAO always means greater useful power.	Power depends on velocity, PTO damping, phase, and constraints. Excess response can increase stroke and loads.
Pitch RAO is dimensionless.	$H_5 = \hat{\theta}/\hat{a}$ is reported in rad/m or deg/m.
Turning off direct pitch moment must eliminate pitch.	Pitch may remain through heave-pitch coupling.
Doubling wave amplitude doubles the RAO.	In a linear model, force and response both double, so their ratio is unchanged.
Every off-diagonal damping entry must be positive.	Individual cross terms can be negative while the full symmetric damping matrix remains positive semidefinite.
The animation is a transient simulation.	It is a reconstruction of the steady harmonic solution at one frequency.

Common Frequency-Domain Theory and Conventions

4.1 Calculation flow

Every control change rebuilds the model and solves it over the full period vector.

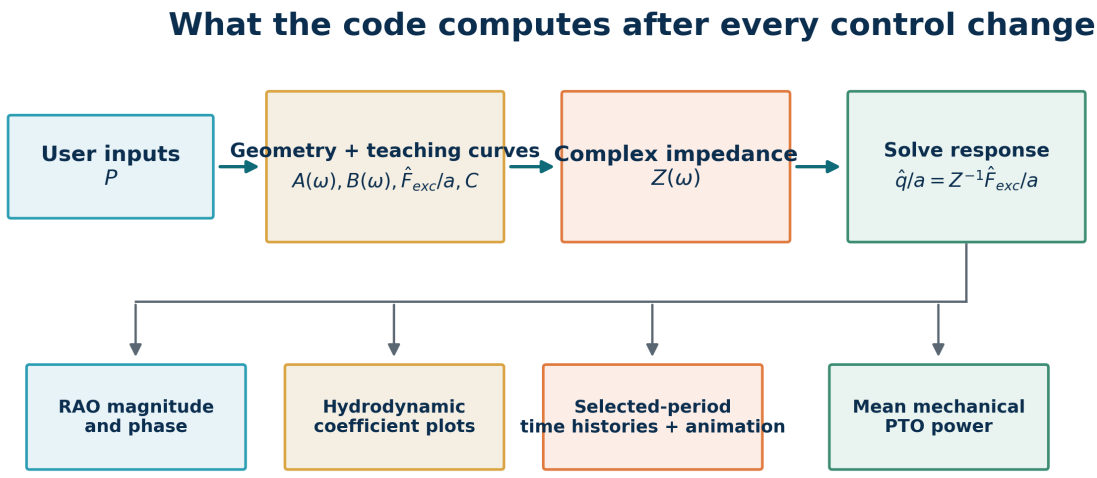


Figure 4.1: Frequency-domain calculation flow shared by both demos.

4.2 Regular-wave and phasor convention

The incident regular wave is represented by

$$\eta(t) = \Re\left\{\hat{a}e^{\mathrm{i}\omega t}\right\}, \tag{4.1}$$

where $\hat{a}$ is the complex wave-elevation amplitude and $a = |\hat{a}|$. A harmonic response is

$$q(t) = \Re\left\{\hat{q}e^{\mathrm{i}\omega t}\right\}, \tag{4.2}$$

so that

$$\dot{q}(t) \leftrightarrow \mathrm{i}\omega\hat{q}, \qquad \ddot{q}(t) \leftrightarrow -\omega^2\hat{q}. \tag{4.3}$$

The use of $e^{\mathrm{i}\omega t}$ determines the sign of the imaginary damping term in the impedance. A data file or external solver using $e^{-\mathrm{i}\omega t}$ can report conjugated phases. Magnitudes may still look correct while phase is reversed, so conventions must be checked before importing complex excitation data.

4.3 Scalar heave equation

For one selected regular-wave frequency,

$$[m + A_{33}(\omega)]\,\ddot{z} + [B_{33}(\omega) + B_{\text{visc}} + B_{\text{PTO}}]\,\dot{z} + [C_{33} + K_{\text{extra}}]\,z = F_{\text{exc},3}(t). \tag{4.4}$$

The associated complex dynamic stiffness is

$$Z_3(\omega) = C_{33} + K_{\text{extra}} - \omega^2\,[m + A_{33}(\omega)] + \mathrm{i}\omega\,[B_{33}(\omega) + B_{\text{visc}} + B_{\text{PTO}}]\,. \tag{4.5}$$

4.4 Coupled heave-pitch equation

Define

$$\boldsymbol{q} = \begin{bmatrix} z \\ \theta \end{bmatrix}, \qquad \boldsymbol{F}_{\text{exc}} = \begin{bmatrix} F_{\text{exc},3} \\ M_{\text{exc},5} \end{bmatrix}. \tag{4.6}$$

The coupled equation is

$$[\boldsymbol{M} + \boldsymbol{A}(\omega)]\,\ddot{\boldsymbol{q}} + [\boldsymbol{B}(\omega) + \boldsymbol{B}_{\text{ext}}]\,\dot{\boldsymbol{q}} + [\boldsymbol{C} + \boldsymbol{K}_{\text{extra}}]\,\boldsymbol{q} = \boldsymbol{F}_{\text{exc}}(t), \tag{4.7}$$

with frequency-domain impedance

$$\boldsymbol{Z}(\omega) = -\omega^2\,[\boldsymbol{M} + \boldsymbol{A}(\omega)] + \mathrm{i}\omega\,[\boldsymbol{B}(\omega) + \boldsymbol{B}_{\text{ext}}] + [\boldsymbol{C} + \boldsymbol{K}_{\text{extra}}]\,. \tag{4.8}$$

The response is found by solving

$$\boldsymbol{Z}(\omega)\hat{\boldsymbol{q}} = \hat{\boldsymbol{F}}_{\text{exc}}. \tag{4.9}$$

The code uses MATLAB's backslash operator rather than explicitly calculating $\boldsymbol{Z}^{-1}$, which is numerically preferable.

4.5 Response amplitude operators

The heave and pitch RAOs are

$$H_3(\omega) = \frac{\hat{z}}{\hat{a}}, \qquad H_5(\omega) = \frac{\hat{\theta}}{\hat{a}}. \tag{4.10}$$

Their units are:

- H_3: m/m, dimensionless;
- H_5: rad/m internally, optionally displayed as deg/m.

The RAO is complex. The demos report magnitude and phase:

$$|H_j| = \frac{|\hat{q}_j|}{|\hat{a}|}, \qquad \angle H_j = \angle \hat{q}_j - \angle \hat{a}. \tag{4.11}$$

A phase jump between approximately $+180°$ and $-180°$ can be a normal wrapping effect, not a physical discontinuity.

4.6 Resonance as an inertia-restoring balance

Ignoring damping for an initial interpretation, a scalar response tends to be large when

$$\omega^2 \left[m + A_{33}(\omega)\right] \approx C_{33} + K_{\text{extra}}. \tag{4.12}$$

For pitch, the analogous diagonal estimate is

$$\omega^2 \left[I_{yy} + A_{55}(\omega)\right] \approx C_{55} + K_{55,\text{extra}}. \tag{4.13}$$

Balance period versus visible RAO maximum

The inertia-restoring balance is an interpretation aid, not a guarantee that the largest RAO occurs at exactly the same period. Frequency-dependent excitation, radiation damping, cross coupling, and the response of the other degree of freedom can move the observed maximum.

4.7 Mean mechanical PTO power

For a linear damper in one harmonic coordinate, the mean absorbed mechanical power is

$$\overline{P}_{\text{PTO}} = \frac{1}{2} B_{\text{PTO}} |\hat{q}|^2 = \frac{1}{2} B_{\text{PTO}} \omega^2 a^2 |H|^2. \tag{4.14}$$

The coupled demo adds the heave and pitch damper contributions:

$$\overline{P}_{\text{total}} = \frac{1}{2} B_{\text{PTO},33} \omega^2 a^2 |H_3|^2 + \frac{1}{2} B_{\text{PTO},55} \omega^2 a^2 |H_5|^2. \tag{4.15}$$

This is ideal mean **mechanical** power dissipated in the modeled linear PTO dampers. It does not include generator efficiency, transmission loss, power electronics, saturation, reactive power, or availability.

4.8 Why wave amplitude does not change a linear RAO

In the linear model,

$$\hat{\boldsymbol{F}}_{\text{exc}} \propto \hat{a}, \qquad \hat{\boldsymbol{q}} = \boldsymbol{Z}^{-1} \hat{\boldsymbol{F}}_{\text{exc}} \propto \hat{a}. \tag{4.16}$$

Therefore $\hat{\boldsymbol{q}}/\hat{a}$ is unchanged. Doubling a doubles the physical motion amplitude and multiplies the ideal damper power by four because power scales with velocity squared.

4.9 Frequency domain versus time domain

The demos solve one steady harmonic response at each frequency. They do not solve the arbitrary-motion radiation-memory problem. A general linear time-domain radiation force is commonly written in Cummins form,

$$\boldsymbol{F}_{\text{rad}}(t) = -\boldsymbol{A}_{\infty} \ddot{\boldsymbol{q}}(t) - \int_0^t \boldsymbol{K}_r(t-\tau) \dot{\boldsymbol{q}}(\tau)\, d\tau. \tag{4.17}$$

The selected-period plots in the GUI evaluate the frequency-domain solution and then reconstruct sinusoidal time histories. They are useful for phase visualization but should not be described as an irregular-wave or transient simulation.

4.10 Physical source of each coefficient

Table 4.1: Typical source of each model input in a device-specific workflow.

Quantity	Typical source
m, I_{yy}	CAD mass properties, weighing, pendulum test, or another physical measurement.
C_{33}, C_{35}, C_{55}	Hydrostatic calculation at a stated equilibrium and rotation reference point.
$A_{ij}(\omega), B_{ij}(\omega)$	Radiation problems from a frequency-domain boundary-element solver.
$F_{\text{exc},3}, M_{\text{exc},5}$	Incident plus diffraction problem; retain magnitude and phase.
PTO damping/stiffness	PTO model, controller design, or linearization about an operating condition.
Mooring terms	Static and dynamic linearization of the mooring model around equilibrium.
Extra viscous damping	Explicit engineering approximation, experiment, or CFD/test calibration.
Wave amplitude and period	Regular-wave test or chosen operating point.

4.11 Units of the coupled matrices

Table 4.2: Units of generalized heave-pitch matrix entries.

Entry	Mass / added mass	Damping	Stiffness
33	kg	N s/m	N/m
35 and 53	kg m	N s	N
55	kg m^2	N m s/rad	N m/rad

Symmetry means $A_{35} = A_{53}$, for example, under consistent reciprocal conventions. The paired entries occupy transpose locations but couple different generalized coordinates, so the dimensional interpretation must include the coordinate and force type.

4.12 Model limitations

The demos assume:

- small-amplitude waves and body motions;
- linear hydrostatics and linear PTO/viscous terms;
- one regular-wave frequency at a time;
- fixed mean geometry and reference point;
- frequency-domain hydrodynamic coefficients;
- no nonlinear drag, separation, slamming, end stops, or saturation; and
- no mooring dynamics beyond optional linear stiffness represented by the extra stiffness controls.

The most valuable interpretation is therefore the *direction and mechanism* of parameter effects, not a claim that a generic device would achieve the plotted response or power.

Demo 1: One-Degree-of-Freedom Heave

5.1 Physical model and teaching purpose

The heave demo represents a generic surface-piercing, axisymmetric point-absorber-type body with one generalized coordinate, $z(t)$. The default reference geometry is a 6 m diameter waterplane and a 4 m displaced-water draft. The dry/body mass is set equal to the reference displaced-water mass at the default condition. The geometry is intentionally simple so that students can calculate A_{wp} and C_{33} by hand before examining frequency-dependent hydrodynamics.

The physical story is:

- body mass and added mass resist acceleration;
- radiation, PTO, and equivalent viscous damping oppose velocity and remove energy;
- hydrostatic and extra stiffness pull the body toward equilibrium; and
- the wave-excitation transfer function drives the response.

5.2 Default parameters and outputs

Table 5.1: Default 1-DOF heave model parameters.

Quantity	Default	Definition or role
Seawater density, ρ	1025 kg/m^3	Internal constant.
Gravity, g	9.81 m/s^2	Internal constant.
Water depth	60 m	Used for the displayed wave number and motion visualization.
Waterplane radius, r	3.0 m	Adjustable geometry input.
Reference hydrodynamic draft, d	4.0 m	Fixed scale used to construct displaced-water mass.
Waterplane area, A_{wp}	28.274 m^2	πr^2.
Body mass, m	115.925 tonne	Default $\rho A_{wp} d$.
Hydrostatic stiffness, C_{33}	284.305 kN/m	$\rho g A_{wp}$.
PTO damping	60 kN s/m	Ideal linear mechanical PTO.
Equivalent viscous damping	20 kN s/m	Additional linear damping correction.
Extra stiffness	0 kN/m	Optional linear PTO/mooring/tuning stiffness.
Wave amplitude, a	0.50 m	Affects physical motion and power, not RAO.
Selected wave period	7.00 s	Used for the marker, time histories, and animation.
Period range	2.5–15 s	520 frequency points.

At default settings, the teaching curves and response have the values in Table 5.2.

Table 5.2: Default 1-DOF teaching-curve and response summary.

Output	Default result over 2.5–15 s
Added mass, A_{33}	74.961–96.654 tonne
Radiation damping, B_{33}	Maximum 119.818 kN s/m
Excitation magnitude per wave amplitude, $\lvert F_{\mathrm{exc},3}\rvert/a$	150.620–326.068 kN/m
Largest heave RAO	$\lvert H_3\rvert = 1.528973$ at $T = 6.667$ s
Estimated diagonal inertia-restoring balance	Reported separately by the interface; it may not exactly match the RAO maximum

5.3 Exact teaching-curve construction

The demo constructs smooth coefficient curves from displaced-water and hydrostatic scales. Define

$$M_d = \rho A_{wp} d, \qquad \omega_c = \frac{2\pi}{6.0\ \mathrm{s}}, \qquad x = \frac{\omega}{\omega_c}. \tag{5.1}$$

The added-mass teaching curve is

$$A_{33}(\omega) = s_A M_d \left[0.62 + \frac{0.18}{1+x^2} + 0.12\exp\left(-\left(\frac{x-1}{0.45}\right)^2\right)\right], \tag{5.2}$$

where s_A is the adjustable added-mass scale.

The radiation-damping curve is

$$B_{33}(\omega) = s_B\, 0.66\sqrt{C_{33}M_d}\, x^2 \exp(1-x^2), \tag{5.3}$$

where s_B is the radiation-damping scale. The form gives $B_{33} \to 0$ at very low and very high frequency and a positive finite radiation band.

The excitation transfer function is real in this 1-DOF teaching model:

$$\frac{F_{\mathrm{exc},3}}{a} = s_F C_{33} \frac{1 + 0.18x^2}{\sqrt{1 + (x/1.55)^6}}, \tag{5.4}$$

where s_F is the excitation scale. This curve is constructed to approach C_{33} for long waves as a pedagogical limiting behavior; it is not a universal identity for every body geometry and wave direction.

Why a geometry spinner is still an approximation

Changing waterplane radius recalculates A_{wp}, displaced-water scale, C_{33}, and the synthetic coefficient magnitudes. In a real geometry study, the complete wetted surface changes, so $A(\omega)$, $B(\omega)$, and complex excitation should be recomputed with a hydrodynamic solver rather than scaled from one baseline.

5.4 Frequency-domain solution

With

$$B_{\mathrm{ext}} = B_{\mathrm{PTO}} + B_{\mathrm{visc}}, \qquad K = C_{33} + K_{\mathrm{extra}}, \tag{5.5}$$

the demo evaluates

$$Z_3(\omega) = K - \omega^2\,[m + A_{33}(\omega)] + \mathrm{i}\omega\,[B_{33}(\omega) + B_{\mathrm{ext}}] \tag{5.6}$$

and

$$H_3(\omega) = \frac{F_{\mathrm{exc},3}/a}{Z_3(\omega)}. \tag{5.7}$$

The selected-period quantities are interpolated with piecewise cubic Hermite interpolation using MATLAB's `interp1(...,'pchip')`.

5.5 Interface map

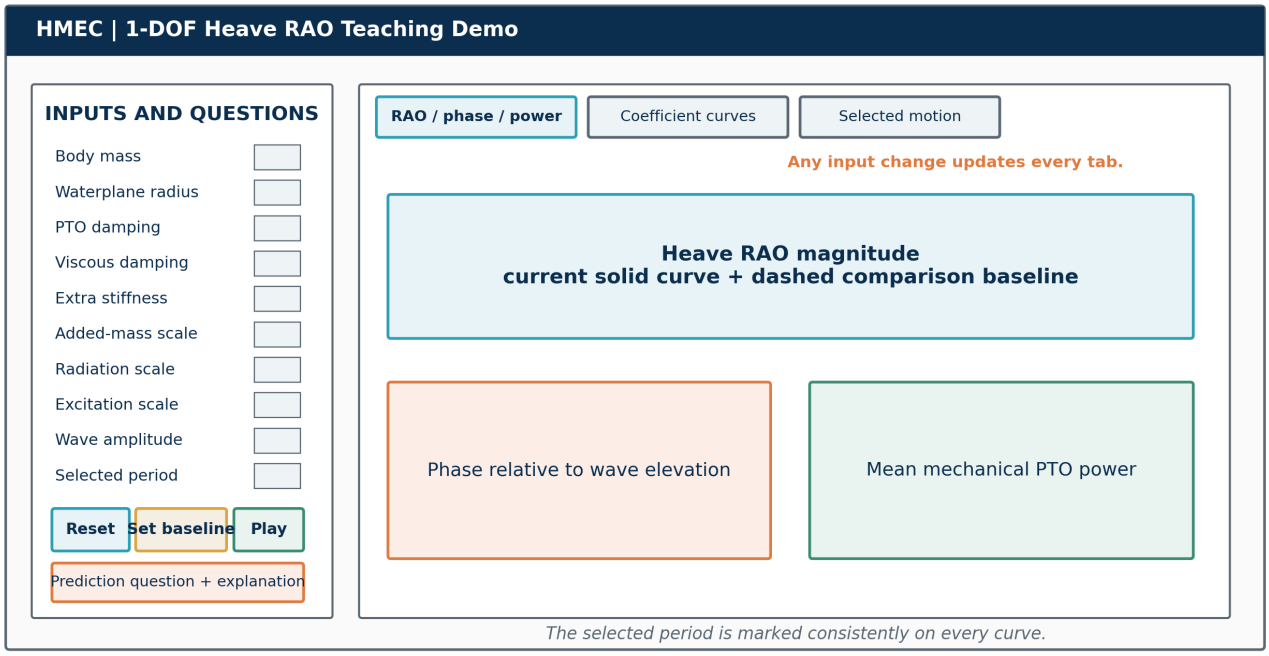


Figure 5.1: Schematic map of the 1-DOF heave interface. The actual GUI contains the same controls, tabs, comparison behavior, selected-period marker, and teaching-question panel.

5.6 Controls

Table 5.3: 1-DOF heave controls, limits, defaults, and primary effects.

Control	Limits	Default	Primary modeled effect
Body mass m [tonne]	45–250	115.925	Changes rigid-body inertia only. Larger mass generally moves the inertia-restoring balance to longer period.
Waterplane radius [m]	1.5–5.0	3.0	Recomputes A_{wp}, C_{33}, displaced-water coefficient scales, and the animation width.
PTO damping [kN s/m]	0–250	60	Adds to external damping and determines ideal mean mechanical PTO power.
Viscous damping [kN s/m]	0–120	20	Adds a frequency-independent equivalent damping correction but does not contribute to displayed PTO power.
Extra stiffness [kN/m]	0–500	0	Adds to hydrostatic restoring; represents a simple linear PTO/mooring/tuning stiffness.
Added-mass scale [-]	0.40–1.80	1.00	Multiplies the full synthetic $A_{33}(\omega)$ curve.
Radiation-damping scale [-]	0.40–1.80	1.00	Multiplies the full synthetic $B_{33}(\omega)$ curve.

Control	Limits	Default	Primary modeled effect
Excitation scale [-]	0.50–1.50	1.00	Multiplies $F_{\mathrm{exc},3}/a$ without changing mass, damping, or stiffness.
Wave amplitude a [m]	0.05–1.50	0.50	Changes physical motion amplitude and ideal damper power; leaves RAO unchanged.
Selected period [s]	2.5–15	7.00	Selects the marker, interpolated values, time histories, dispersion wave number, and animation.

5.7 Buttons and status indicator

- **Reset**: restores every input to its default value and clears the comparison state.
- **Set baseline**: stores the current response for dashed-curve comparison.
- **Play motion**: animates several cycles of the selected harmonic wave and heave response.
- **Reveal explanation**: displays the model-based explanation for the selected prediction question.
- **Status**: reports finite-response, coefficient-sign, positive-stiffness, and selected-period checks.

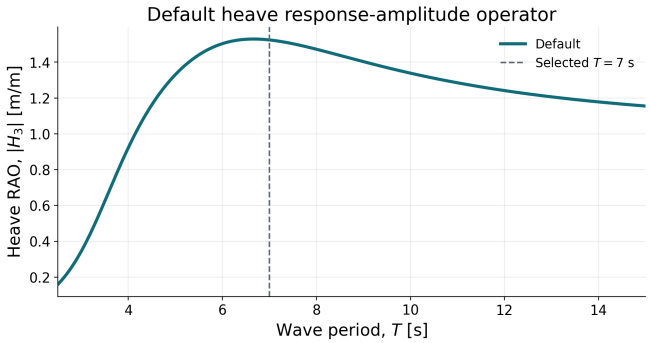
5.8 Tabs and plot interpretation

5.8.1 RAO, phase, and PTO power

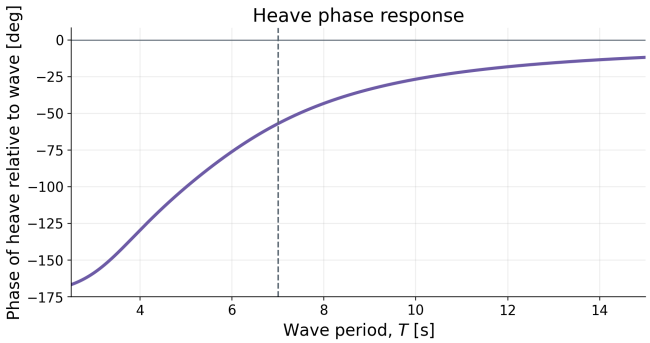
This tab shows:

- $|H_3|$ versus wave period;
- phase of heave relative to wave elevation; and
- ideal mean mechanical PTO power from Equation (4.14).

The solid curve is the current case. After **Set baseline**, a dashed curve shows the stored comparison. A vertical dotted marker identifies the selected period.



(a) Default RAO magnitude.



(b) Default phase.

Figure 5.2: Default heave response magnitude and phase generated from the same formulas as the MATLAB demo.

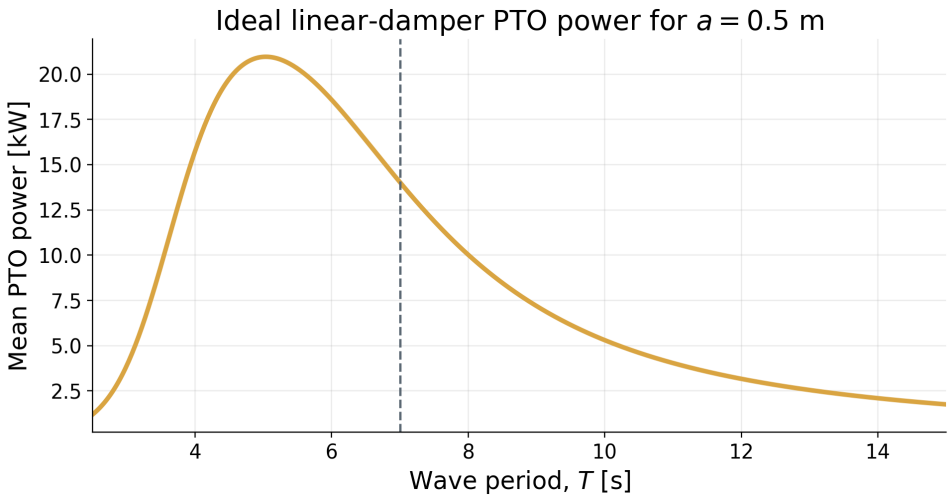
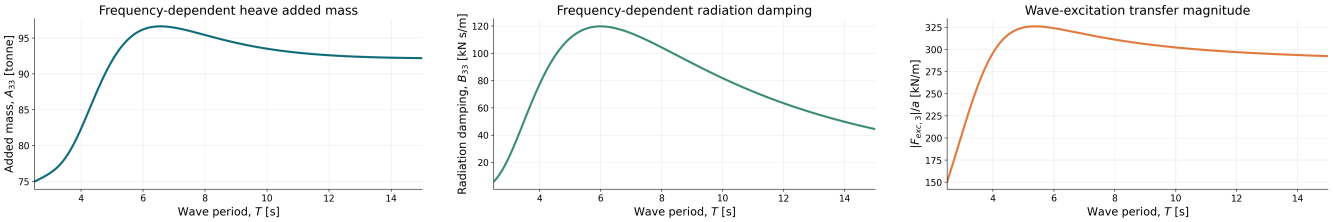


Figure 5.3: Default ideal mean mechanical PTO-power curve. The plotted quantity is not net electrical power.

5.8.2 Hydrodynamic coefficient curves

This tab shows A_{33}, B_{33}, and $|F_{\mathrm{exc},3}|/a$ in physical units. The coefficient plots are important because two cases can have similar RAOs for different physical reasons.



(a) Added mass. **(b)** Radiation damping. **(c)** Excitation per wave amplitude.

Figure 5.4: Default 1-DOF synthetic teaching coefficients.

5.8.3 Selected regular-wave response

This tab contains:

- a schematic wave-body animation;
- wave elevation and heave time histories reconstructed over several cycles; and
- a text summary of selected-period frequency, wave number, coefficient values, RAO magnitude/phase, physical response amplitude, and ideal PTO power.

The finite-depth dispersion relation is solved numerically:

$$\omega^2 = gk \tanh(kh). \tag{5.8}$$

The wave number affects the visual wavelength; it does not modify the preconstructed hydrodynamic teaching curves.

5.9 Built-in prediction experiments

5.9.1 Experiment H1: increase body mass

Prediction

If body mass increases while hydrostatic stiffness and hydrodynamic curves remain fixed, should the main response region move toward shorter or longer period?

Procedure:

1. Reset the demo and click **Set baseline**.
2. Increase body mass by about 40%, from 115.9 to approximately 162 tonne.
3. Compare RAO magnitude, phase, and the estimated balance period.

Expected result: greater inertia requires a smaller angular frequency for the inertia-restoring balance, so the response region moves to longer period. The independent check moves the RAO peak from 6.667 s to 7.124 s.

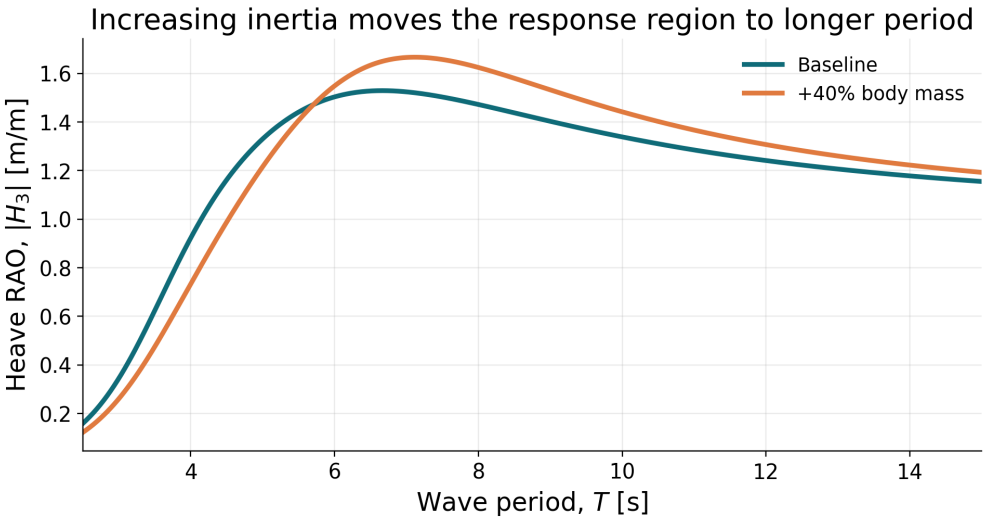


Figure 5.5: Effect of increasing body mass in the 1-DOF teaching model.

5.9.2 Experiment H2: increase added mass

The interpretation is similar to body mass because $m + A_{33}$ forms the effective inertia. Unlike body mass, added mass is a frequency-dependent hydrodynamic reaction. Scaling the entire curve isolates its inertial role but is not equivalent to changing geometry in a real BEM model.

5.9.3 Experiment H3: increase waterplane radius

Increasing radius produces a strong r^2 increase in A_{wp} and C_{33}. In the present synthetic scaling, stiffness tends to dominate the period shift and the response region moves shorter. Students should be asked why a real geometry change also requires recomputing diffraction and radiation.

5.9.4 Experiment H4: increase PTO damping

Prediction

Will increasing PTO damping always increase absorbed power? What will it do to the RAO peak?

Expected result: higher damping lowers and broadens the response peak. Power is a product of damping and velocity squared; it can increase initially but ultimately falls when the motion is suppressed. Therefore, the largest allowable damping is not automatically the optimal damping.

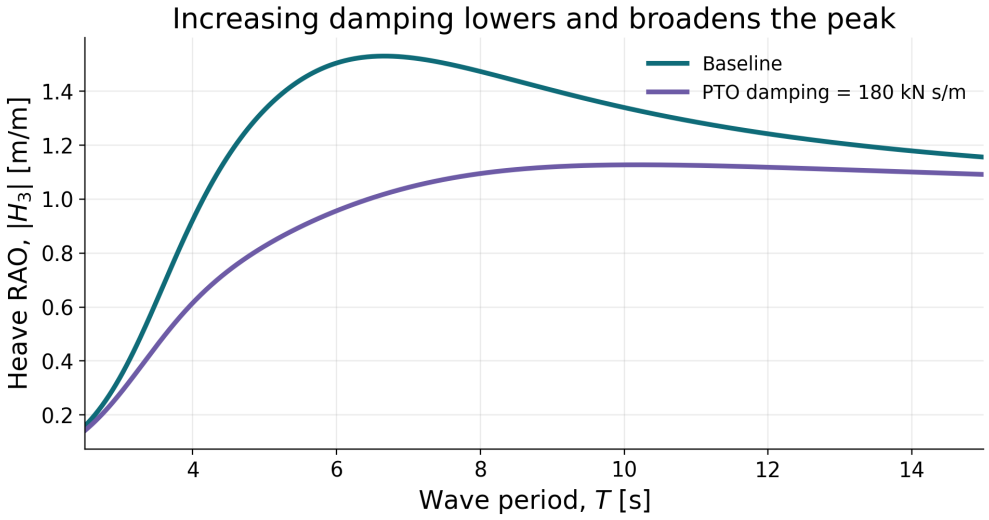


Figure 5.6: Higher PTO damping lowers and broadens the heave RAO peak in the teaching model.

5.9.5 Experiment H5: increase extra stiffness

Increasing K_{extra} increases restoring while inertia is unchanged. The response region moves to shorter period. The independent test adds 200 kN/m and moves the RAO peak from 6.667 s to 4.234 s.

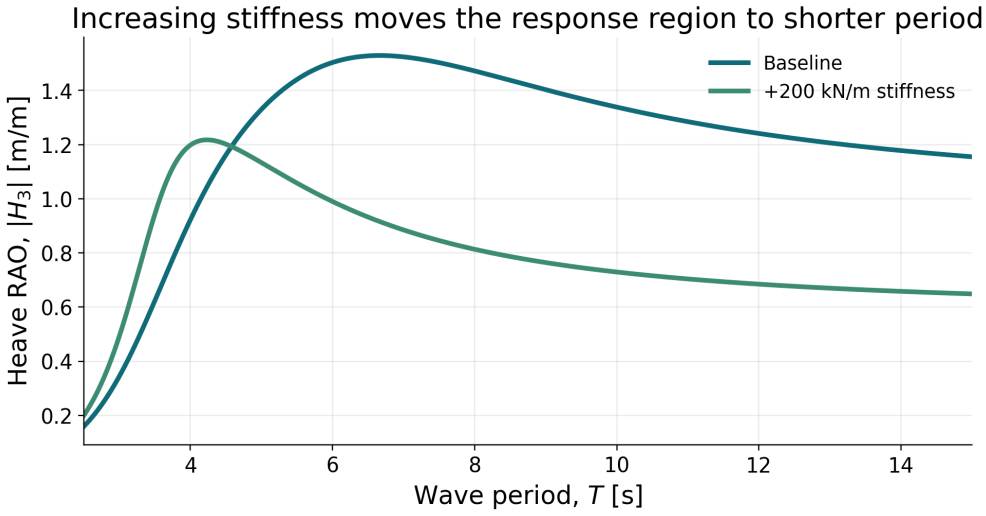


Figure 5.7: Effect of adding 200 kN/m of extra linear stiffness.

5.9.6 Experiment H6: double wave amplitude

Prediction

If wave amplitude doubles from 0.50 m to 1.00 m, which of the following change: RAO, motion amplitude, phase, or ideal damper power?

Expected result:

- RAO magnitude and phase are unchanged;
- physical heave amplitude doubles; and
- ideal mean PTO power increases by a factor of four.

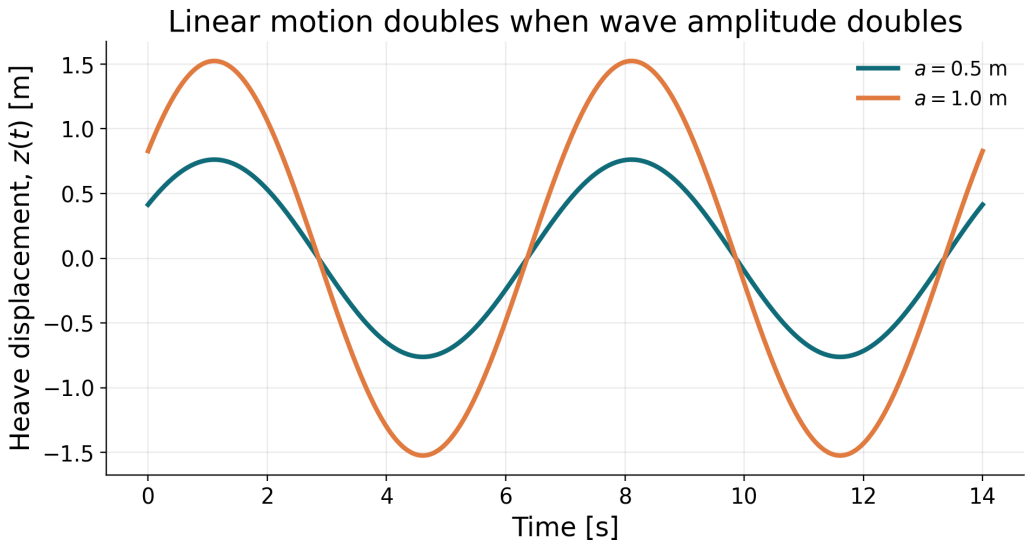


Figure 5.8: Wave-amplitude change affects physical motion but not the linear RAO.

5.10 Suggested discussion of phase

Below the main resonance region, restoring tends to dominate. Around resonance, the response passes through a damping-controlled phase relation. Above resonance, inertia dominates and the motion approaches an opposite-phase response. The exact phase curve also includes the phase of excitation. In this 1-DOF teaching model the excitation transfer is real and positive, simplifying the interpretation.

5.11 Diagnostics and self-test logic

The GUI status check verifies that:

- all RAO values are finite;
- added mass is positive;
- radiation damping is nonnegative;
- total stiffness is positive; and

- selected period lies inside the solved range.

The embedded self-test additionally checks the mass, stiffness, damping, and wave-amplitude trends. A passing self-test confirms internal consistency of the teaching model, not agreement with a real device.

5.12 Key teaching messages from the heave demo

End-of-demo summary

1. The RAO is the result of a competition among inertia, damping, restoring, and excitation.
2. Added mass and radiation damping are hydrodynamic and frequency dependent.
3. Damping changes peak height and bandwidth; stiffness and inertia strongly affect peak location.
4. The largest motion is not automatically the largest useful power.
5. In a linear model, wave amplitude scales motion and power but not the RAO.

Demo 2: Coupled Heave and Pitch

6.1 Physical model and teaching purpose

The coupled demo represents a generic elongated 18 m floating WEC or small hull with two generalized coordinates,

$$\boldsymbol{q} = \begin{bmatrix} z \\ \theta \end{bmatrix}. \tag{6.1}$$

It is not a detailed hull definition. The reference length, beam, waterplane area, mass, equilibrium draft, and pitch inertia establish realistic scales for the teaching matrices and animation.

The central new idea is that each equation can contain reactions caused by both coordinates. For example, $A_{35}\ddot{\theta}$ is a vertical force associated with pitch acceleration, while $A_{53}\ddot{z}$ is a pitch moment associated with heave acceleration. Under consistent reciprocal conventions, the coefficient matrices are symmetric, but their entries retain different generalized units.

6.2 Default parameters and outputs

Table 6.1: Default coupled heave-pitch model parameters.

Quantity	Default	Definition or role
Seawater density, ρ	1025 kg/m^3	Internal constant.
Gravity, g	9.81 m/s^2	Internal constant.
Water depth	80 m	Used for wave number and motion visualization.
Reference length	18.0 m	Used for pitch scale, hydrodynamic inertia, animation, and scaled-impedance display.
Reference beam	6.0 m	Used for the schematic body.
Waterplane area, A_{wp}	80.0 m^2	Adjustable heave hydrostatic input.
Body mass, m	420 tonne	Adjustable rigid-body mass.
Reference equilibrium draft	5.122 m	$m/(\rho A_{wp})$ at default settings.
Pitch inertia, I_{yy}	13.0×10^6 kg m^2	Adjustable rigid-body pitch inertia.
Heave restoring, C_{33}	804.420 kN/m	$\rho g A_{wp}$.
Pitch restoring, C_{55}	18.0 MN m/rad	Adjustable hydrostatic/tuning pitch stiffness.
Hydrostatic cross term, $C_{35} = C_{53}$	380.520 kN	Default 10% correlation construction.
Heave PTO damping	130 kN s/m	Ideal heave mechanical PTO.
Pitch PTO damping	2.50 MN m s/rad	Ideal pitch mechanical PTO.
Extra heave damping	30 kN s/m	Equivalent viscous correction.
Extra pitch damping	0.60 MN m s/rad	Equivalent viscous correction.
Wave amplitude, a	0.50 m	Affects physical motion and power, not RAOs.
Selected period	7.00 s	Marker, interpolation, time histories, and animation.
Period range	3–16 s	560 frequency points.

Table 6.2: Default coupled teaching-curve and response summary.

Output	Default result over 3–16 s
A_{33}	230.852–329.408 tonne
A_{55}	7.559 – –9.863 $\times 10^6$ kg m^2
Maximum B_{33}	319.689 kN s/m
Maximum B_{55}	5.199 MN m s/rad
Largest heave RAO	$\lvert H_3 \rvert$ = 1.949580 m/m at T = 6.698 s
Largest pitch RAO	$\lvert H_5 \rvert$ = 0.170528 rad/m = 9.770510 deg/m at T = 6.349 s

6.3 Rigid-body, restoring, and external damping matrices

The body mass matrix is diagonal about the chosen pitch reference point:

$$\boldsymbol{M} = \begin{bmatrix} m & 0 \\ 0 & I_{yy} \end{bmatrix}. \tag{6.2}$$

The default hydrostatic matrix is

$$\boldsymbol{C} = \begin{bmatrix} C_{33} & C_{35} \\ C_{35} & C_{55} \end{bmatrix}, \qquad C_{33} = \rho g A_{wp}. \tag{6.3}$$

The coupling spinner is converted to $c = P_{\text{coupling}}/100$, and the cross term is constructed as

$$C_{35} = C_{53} = 0.10c\sqrt{C_{33}C_{55}}. \tag{6.4}$$

This correlation form preserves a positive restoring matrix over the classroom range while allowing the equations to be decoupled at $c = 0$.

Extra stiffness is diagonal:

$$\boldsymbol{K}_{\text{extra}} = \begin{bmatrix} K_{33,\text{extra}} & 0 \\ 0 & K_{55,\text{extra}} \end{bmatrix}. \tag{6.5}$$

The external damping matrix is also diagonal:

$$\boldsymbol{B}_{\text{ext}} = \begin{bmatrix} B_{\text{PTO},33} + B_{\text{visc},33} & 0 \\ 0 & B_{\text{PTO},55} + B_{\text{visc},55} \end{bmatrix}. \tag{6.6}$$

The teaching package does not include off-diagonal PTO damping, although a real multi-body PTO or controller could introduce it.

6.4 Exact added-mass teaching matrix

The reference displaced mass is

$$M_d = \rho A_{wp} d_{\text{ref}}, \tag{6.7}$$

and the reference hydrodynamic pitch-inertia scale is

$$I_{h,\text{ref}} = \frac{M_d\left(L^2 + d_{\text{ref}}^2\right)}{12}. \tag{6.8}$$

Define

$$x_3 = \frac{\omega}{2\pi/(7.2\ \text{s})}, \qquad x_5 = \frac{\omega}{2\pi/(8.8\ \text{s})}. \tag{6.9}$$

Then

$$A_{33} = s_A M_d \left[0.52 + \frac{0.20}{1 + x_3^2} + 0.16\exp\left(-\left(\frac{x_3 - 1}{0.50}\right)^2\right)\right], \tag{6.10}$$

$$A_{55} = s_A I_{h,\text{ref}} \left[0.60 + \frac{0.16}{1 + x_5^2} + 0.12\exp\left(-\left(\frac{x_5 - 1}{0.55}\right)^2\right)\right]. \tag{6.11}$$

The cross added mass is

$$A_{35} = A_{53} = r_A(\omega)\sqrt{A_{33}A_{55}}, \tag{6.12}$$

where

$$r_A(\omega) = 0.18c \exp\left[-\left(\frac{0.5(x_3 + x_5) - 1}{1.40}\right)^2\right]. \tag{6.13}$$

The correlation construction gives a symmetric positive-semidefinite matrix while allowing a visually meaningful coupling trend.

6.5 Exact radiation-damping teaching matrix

The diagonal curves are

$$B_{33} = s_B\, 0.55\sqrt{C_{33}M_d}\, x_3^2 \exp(1 - x_3^2), \tag{6.14}$$

$$B_{55} = s_B\, 0.35\sqrt{C_{55}I_{h,\mathrm{ref}}}\, x_5^2 \exp(1 - x_5^2). \tag{6.15}$$

The cross term is

$$B_{35} = B_{53} = r_B(\omega)\sqrt{B_{33}B_{55}}, \tag{6.16}$$

with

$$r_B(\omega) = -0.22c \exp\left[-\left(\frac{0.5(x_3 + x_5) - 1}{1.20}\right)^2\right]. \tag{6.17}$$

Why the cross damping can be negative

A negative B_{35} does not imply that the system creates energy. Passivity depends on the quadratic form $\dot{\boldsymbol{q}}^{\mathsf{T}}\boldsymbol{B}\dot{\boldsymbol{q}}$ and therefore on the eigenvalues or positive-semidefinite character of the complete matrix. The self-test checks the full matrix, not the sign of one cross term.

6.6 Complex excitation force and moment

The finite-depth dispersion relation in Equation (5.8) supplies wave number k. Define

$$S_3(\omega) = \frac{1 + 0.12x_3^2}{\sqrt{1 + (x_3/1.65)^6}}, \tag{6.18}$$

$$S_5(\omega) = \frac{\exp(-0.10x_5^2)}{\sqrt{1 + (x_5/1.80)^4}}, \tag{6.19}$$

$$\widehat{\frac{\partial\eta/\partial x}{a}} = -\mathrm{i}kS_5. \tag{6.20}$$

The excitation vector per unit incident-wave amplitude is constructed as

$$\frac{\hat{F}_{\mathrm{exc},3}}{a} = C_{33}S_3 + 0.25C_{35}\,(-\mathrm{i}kS_5)\,, \tag{6.21}$$

$$\frac{\hat{M}_{\mathrm{exc},5}}{a} = C_{35}S_3 + C_{55}\,(-\mathrm{i}kS_5)\,. \tag{6.22}$$

The complex wave-slope term gives force and moment different phases. Clearing **Include direct wave-excitation pitch moment** sets only $\hat{M}_{\mathrm{exc},5}$ to zero; it does not remove hydrodynamic or hydrostatic coupling.

Teaching excitation is not a substitute for BEM diffraction data

The excitation model in Equation (6.22) is designed to retain complex phase and realistic scaling. A device-specific model must import the complex excitation force and moment from the same geometry, water depth, wave direction, reference point, and harmonic convention as the radiation coefficients.

6.7 Coupled frequency-domain solution

At every frequency, the code assembles

$$\boldsymbol{Z} = -\omega^2(\boldsymbol{M} + \boldsymbol{A}) + \mathrm{i}\omega(\boldsymbol{B} + \boldsymbol{B}_{\mathrm{ext}}) + (\boldsymbol{C} + \boldsymbol{K}_{\mathrm{extra}}) \tag{6.23}$$

and solves

$$\begin{bmatrix} H_3 \\ H_5 \end{bmatrix} = \boldsymbol{Z}^{-1} \frac{\hat{\boldsymbol{F}}_{\mathrm{exc}}}{a} \tag{6.24}$$

using MATLAB's matrix left division. The code stores the complete 2×2 matrices over the 560-period grid.

6.8 Interface map

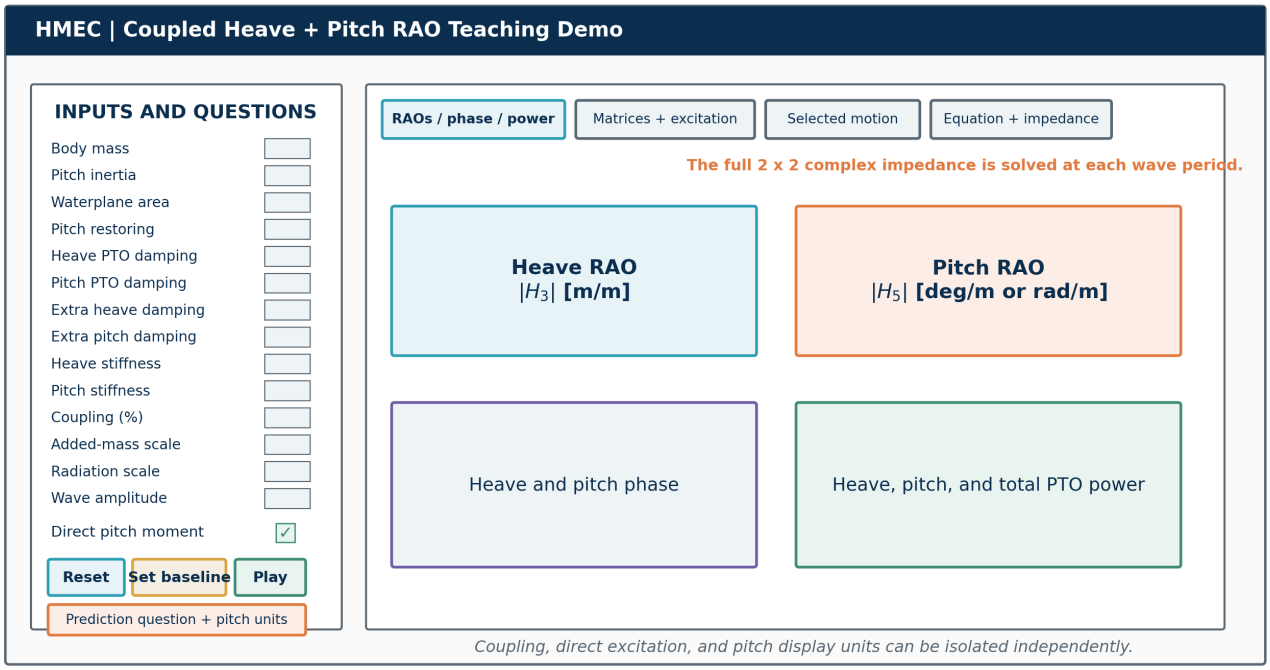


Figure 6.1: Schematic map of the coupled heave-pitch interface. The actual GUI includes four tabs, a direct pitch-excitation checkbox, pitch-unit selection, and matrix diagnostics.

6.9 Controls

Table 6.3: Coupled heave-pitch controls, limits, defaults, and primary effects.

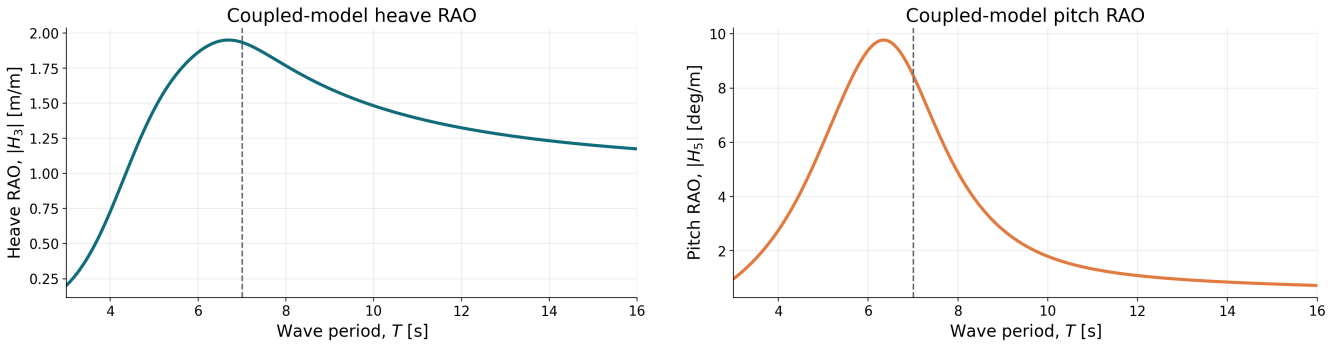
Control	Limits	Default	Primary modeled effect
Body mass [tonne]	150–1000	420	Changes rigid-body heave inertia and the reference equilibrium draft.
Pitch inertia [10^6 kg m^2]	3–60	13	Changes rigid-body pitch inertia only.
Waterplane area [m^2]	25–180	80	Recomputes C_{33}, reference draft, displaced mass scale, and coupling scales.
Pitch restoring [MN m/rad]	3–80	18	Changes C_{55}, hydrostatic cross scale, and pitch damping scale.
Heave PTO damping [kN s/m]	0–600	130	Adds heave damping and determines heave PTO-power contribution.
Pitch PTO damping [MN m s/rad]	0–20	2.50	Adds pitch damping and determines pitch PTO-power contribution.
Extra heave damping [kN s/m]	0–250	30	Equivalent heave viscous correction; excluded from PTO power.
Extra pitch damping [MN m s/rad]	0–10	0.60	Equivalent pitch viscous correction; excluded from PTO power.
Extra heave stiffness [kN/m]	0–1200	0	Adds to C_{33} on the diagonal.
Extra pitch stiffness [MN m/rad]	0–80	0	Adds to C_{55} on the diagonal.
Heave-pitch coupling [%]	0–150	100	Scales C_{35}, A_{35}, and B_{35} correlation terms.
Added-mass scale [-]	0.40–1.80	1.00	Multiplies all added-mass entries through the diagonal scales and cross correlation.
Radiation-damping scale [-]	0.40–1.80	1.00	Multiplies all radiation-damping entries.
Wave amplitude [m]	0.05–1.50	0.50	Changes physical response and ideal power; leaves both RAOs unchanged.
Direct pitch-excitation checkbox	on/off	on	When off, sets $M_{\mathrm{exc},5} = 0$ but leaves coupling active.
Selected period [s]	3–16	7.00	Selects marker, interpolation, animation, and impedance breakdown.
Pitch display units	deg/m or rad/m	deg/m	Changes display conversion only; internal calculations remain in radians.

6.10 Tabs and plot interpretation

6.10.1 Coupled RAOs, phase, and power

The first tab shows four outputs:

- heave RAO magnitude $|H_3|$ in m/m;
- pitch RAO magnitude $|H_5|$ in the selected units;
- heave and pitch phases relative to wave elevation; and
- separate heave, pitch, and total ideal mean PTO power.



(a) Default heave RAO.

(b) Default pitch RAO.

Figure 6.2: Default coupled response magnitudes. Pitch is shown in both rad/m and deg/m in the generated teaching figure.

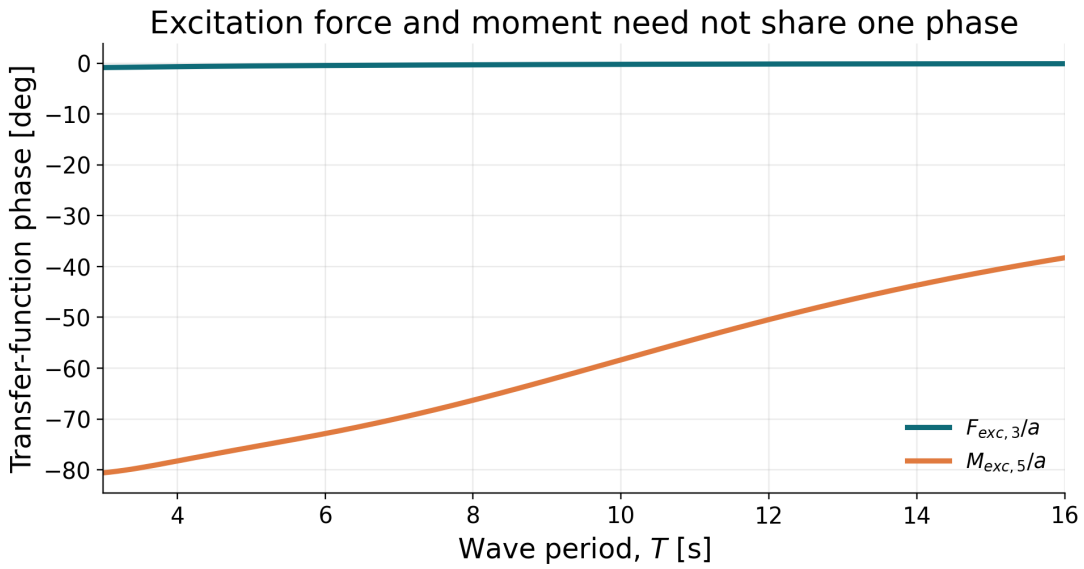
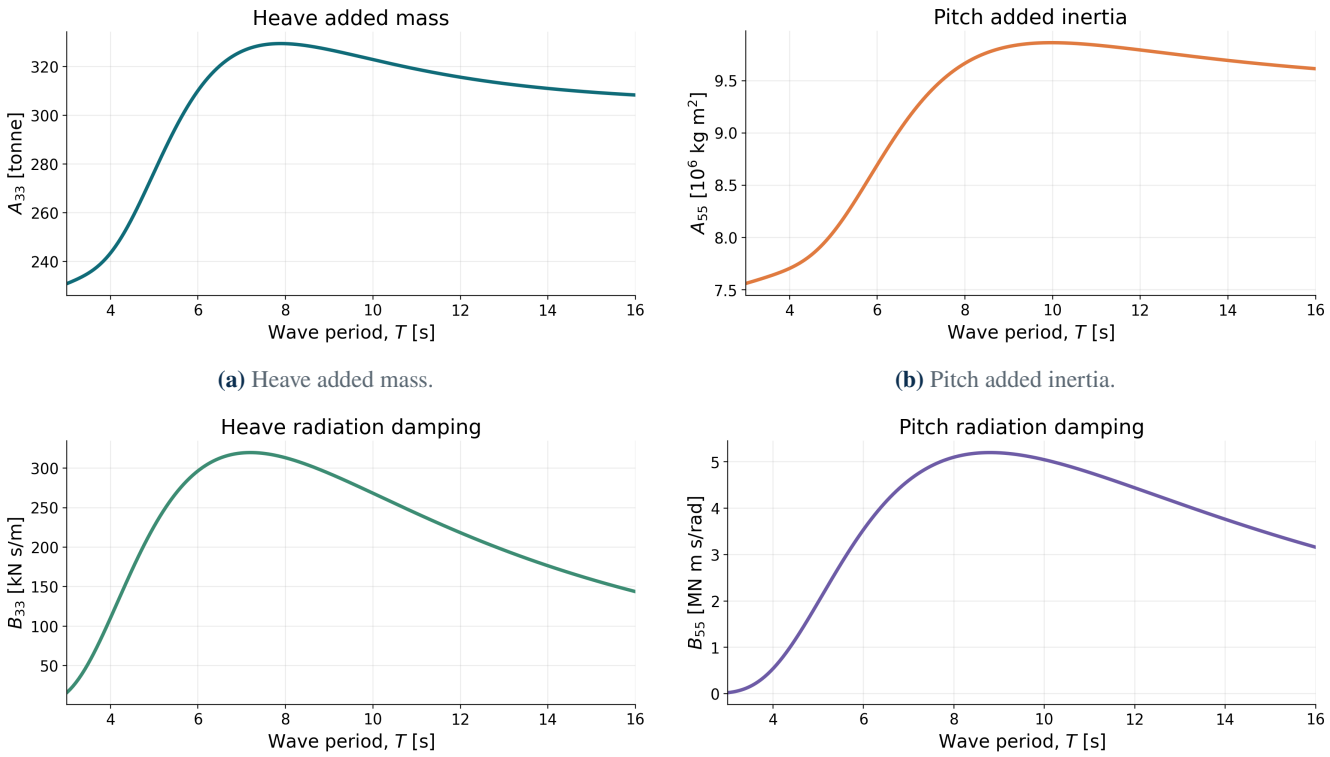


Figure 6.3: Complex excitation phase and its relation to the coupled response. Retaining phase is essential when importing BEM data.

6.10.2 Hydrodynamic matrices and excitation

The second tab plots:

- A_{33}, A_{55}, and $A_{35} = A_{53}$;
- B_{33}, B_{55}, and $B_{35} = B_{53}$; and
- magnitudes and phases of heave excitation force and pitch excitation moment per wave amplitude.



(a) Heave added mass.

(b) Pitch added inertia.

(c) Heave radiation damping.

(d) Pitch radiation damping.

Figure 6.4: Selected diagonal hydrodynamic coefficient curves in the coupled demo.

6.10.3 Selected regular-wave motion

The third tab shows simultaneous heave and pitch motion. The body reference point follows

$$z(t) = \Re\{aH_3 e^{i\omega t}\}, \qquad \theta(t) = \Re\{aH_5 e^{i\omega t}\}. \tag{6.25}$$

The animation exaggerates neither coordinate intentionally, but the schematic body is not a CAD-scale hydrostatic geometry. Use the numerical time histories and summary values for quantitative interpretation.

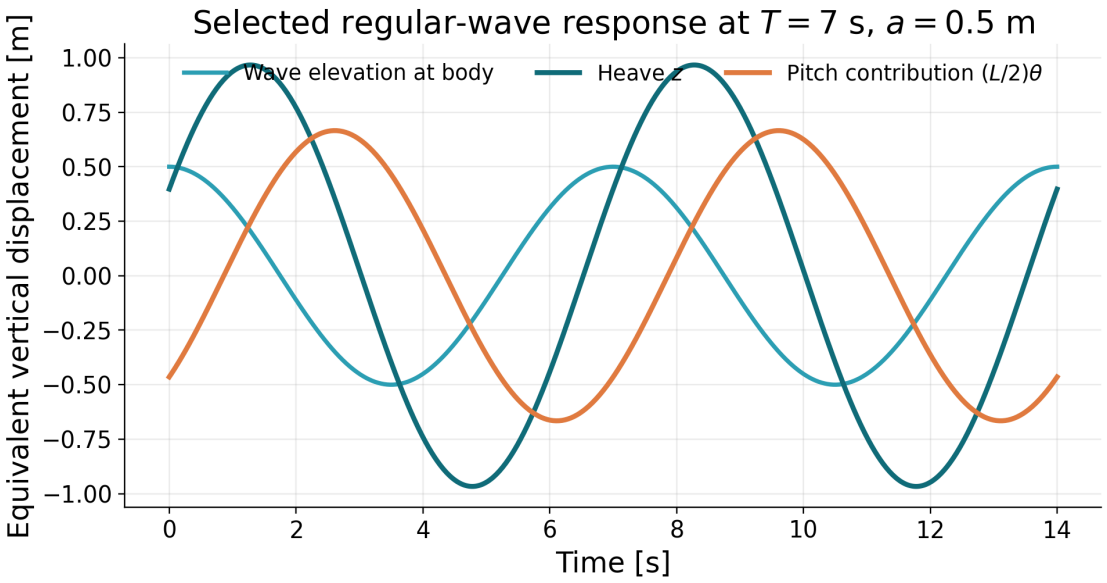


Figure 6.5: Representative selected-period heave and pitch time histories generated by the coupled teaching model.

6.10.4 Equation, units, and impedance

The fourth tab provides:

- the expanded and matrix equations;
- translation-rotation units;
- selected-period effective mass, damping, stiffness, and total impedance; and
- an impedance-contribution plot after scaling pitch as an equivalent length $L\theta$.

The equivalent coordinate is

$$\boldsymbol{q}_s = \begin{bmatrix} z \\ L\theta \end{bmatrix}, \tag{6.26}$$

and the second generalized force is scaled as M_5/L. This makes the displayed impedance entries comparable in kN/m. It is a visualization transformation, not a change to the internally solved physical equations.

6.11 Built-in prediction experiments

6.11.1 Experiment C1: remove all coupling

Set coupling to 0%. The off-diagonal entries A_{35}, B_{35}, and C_{35} vanish. Heave and pitch then solve as two independent scalar equations. If direct pitch excitation remains on, pitch can still respond to $M_{\mathrm{exc},5}$; decoupled does not mean unforced.

6.11.2 Experiment C2: turn off the direct pitch moment

Prediction

If $M_{exc,5} = 0$, must H_5 also be zero?

With coupling active, the answer is no. A vertical excitation force produces heave, and the off-diagonal impedance transfers part of that response into pitch. Pitch becomes exactly zero only when the direct pitch moment is off *and* the coupling terms are zero in this two-DOF model.

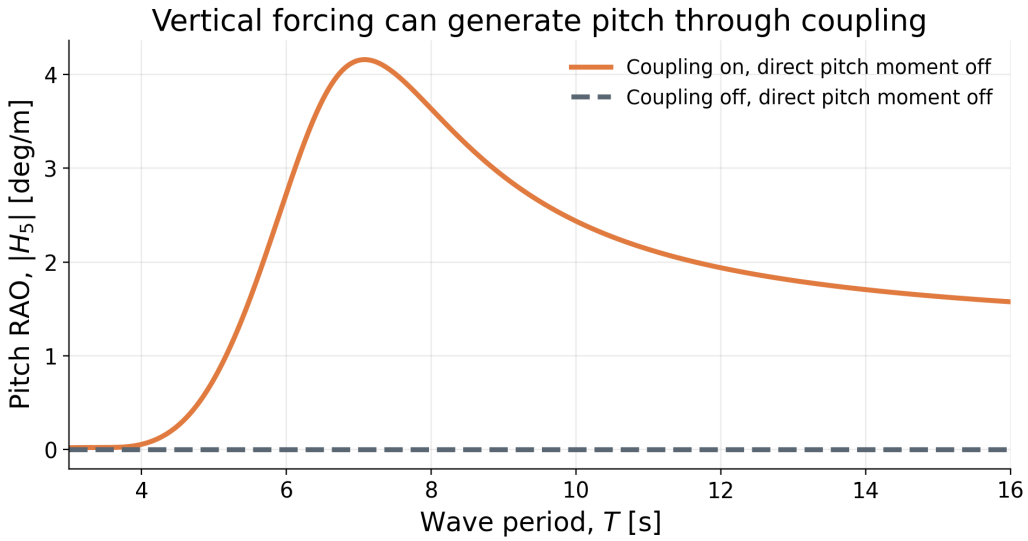


Figure 6.6: Pitch generated by coupling when the direct excitation moment is removed.

6.11.3 Experiment C3: increase pitch inertia

Increase I_{yy} by 50%. The pitch-dominated inertia-restoring balance moves from approximately 7.0 s to 8.0 s in the independent check. The exact pitch RAO maximum also depends on excitation, damping, and coupling.

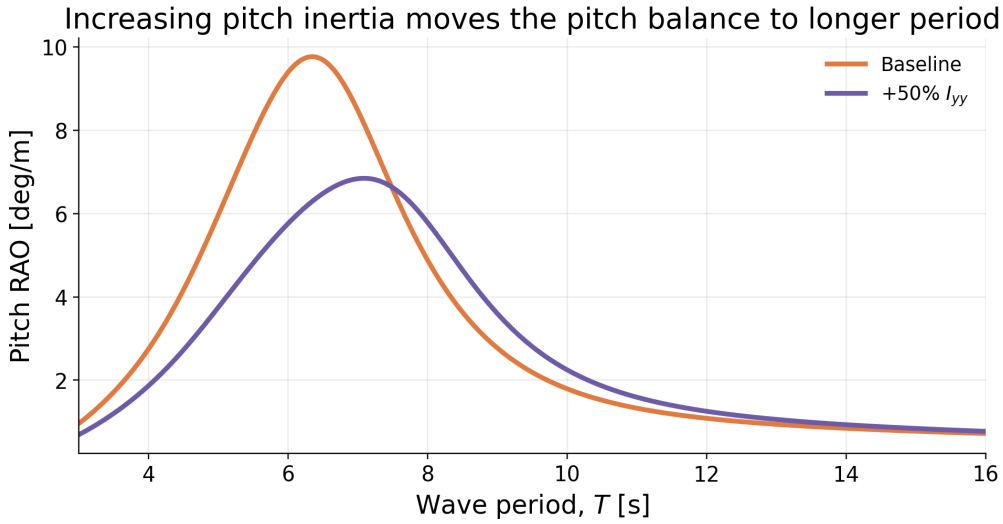


Figure 6.7: Increasing pitch inertia moves the pitch response toward longer period.

6.11.4 Experiment C4: increase pitch stiffness

Increasing C_{55} or extra pitch stiffness moves the pitch-dominated balance toward shorter period. It also changes the hydrostatic cross scale and the synthetic pitch-damping scale in this teaching construction, so students should inspect both the RAO and coefficient tabs.

6.11.5 Experiment C5: increase pitch PTO damping

Higher pitch PTO damping lowers and broadens the pitch peak. The independent high-damping case reduces the largest pitch RAO from 0.170528 to 0.105767 rad/m. As in heave, the power optimum is a balance between larger damping and smaller velocity.

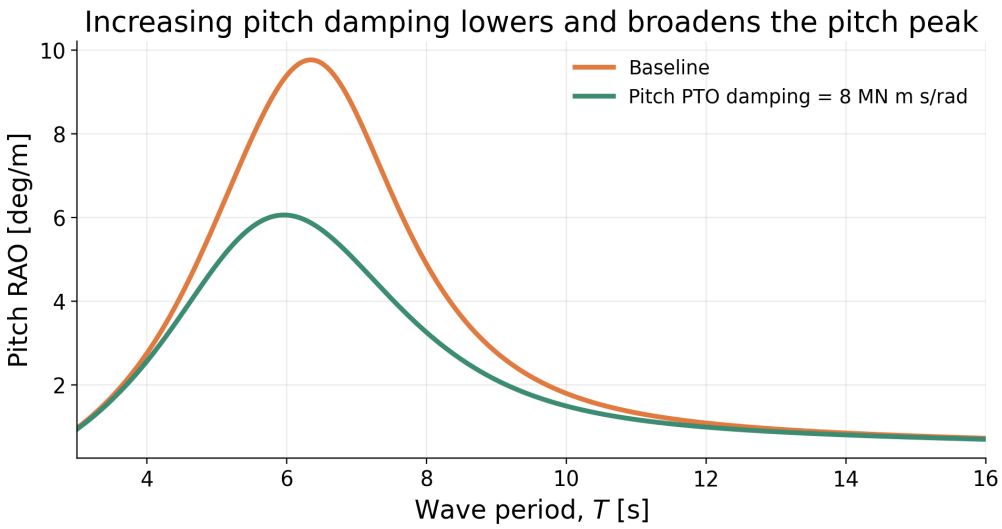


Figure 6.8: Effect of increased pitch PTO damping.

6.11.6 Experiment C6: increase heave stiffness in the coupled system

Increasing heave stiffness changes heave directly and can change pitch indirectly. The diagonal heave balance moves from approximately 5.977 s to 4.349 s after adding 600 kN/m in the independent test.

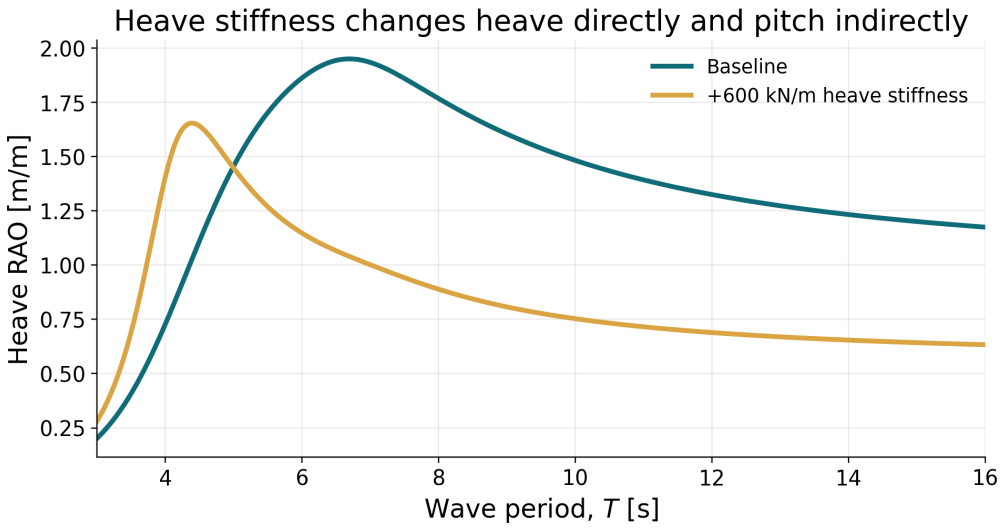


Figure 6.9: A heave-stiffness change modifies both direct heave response and coupling-mediated pitch behavior.

6.11.7 Experiment C7: double wave amplitude

Both RAOs remain unchanged. The heave and pitch motion amplitudes double, and each ideal damper-power contribution increases by a factor of four.

6.11.8 Experiment C8: change pitch display units

Switch from deg/m to rad/m. The plotted numerical values change by $180/\pi$, but the internal matrices, response solution, physical angle, and power do not change. This is a useful reminder that radians are used internally even when the visual display uses degrees.

6.12 Matrix diagnostics and self-test logic

The GUI checks:

- finite RAOs;
- symmetry of $\boldsymbol{A}$, $\boldsymbol{B}$, and $\boldsymbol{C}$;
- positive-semidefinite added-mass and radiation-damping matrices;
- positive total restoring stiffness; and
- a valid selected period.

The self-test also verifies that:

- zero coupling and zero direct pitch moment give exactly zero pitch;
- coupling alone can transfer heave forcing into pitch;
- increased pitch inertia moves the balance longer;
- increased pitch damping lowers the pitch peak;

- increased heave stiffness moves the heave balance shorter; and
- doubling wave amplitude changes neither RAO.

6.13 Key teaching messages from the coupled demo

End-of-demo summary

1. A scalar RAO denominator becomes a complex matrix impedance.
2. Off-diagonal terms mean that one motion can create a force or moment in another equation.
3. Force and moment excitation are complex and can have different phases.
4. Removing direct pitch excitation does not necessarily remove pitch response.
5. Matrix symmetry, positive-semidefinite damping, units, sign conventions, and reference point must be checked together.
6. Pitch RAO is an angular response per unit wave amplitude, not a dimensionless number.

Replacing Teaching Curves with BEM or Experimental Data

7.1 When to replace the synthetic curves

The built-in curves are appropriate for teaching the response equation. Replace them when the goal becomes device-specific prediction, comparison with experiment, WEC-Sim input preparation, control design using a defined geometry, or quantitative performance assessment.

A linear potential-flow BEM typically supplies:

- frequency-dependent added-mass matrix $\boldsymbol{A}(\omega)$;
- frequency-dependent radiation-damping matrix $\boldsymbol{B}(\omega)$;
- complex excitation-force and excitation-moment vector per unit wave amplitude; and
- hydrostatic restoring coefficients, depending on the solver and workflow.

Capytaine, WAMIT, Nemoh, AQWA, and similar tools can support this workflow. WEC-Sim subsequently uses hydrodynamic coefficients from a BEM data structure in its time-domain equations.

7.2 Minimum data required for the heave demo

For a device-specific heave calculation, provide all quantities on a common angular-frequency vector:

$$\omega_j, \quad A_{33}(\omega_j), \quad B_{33}(\omega_j), \quad \frac{\hat{F}_{\mathrm{exc},3}(\omega_j)}{a}. \tag{7.1}$$

The excitation transfer must remain complex when phase is available. The current 1-DOF teaching demo uses a real excitation curve; replacing it with complex data improves the physical phase interpretation.

Table 7.1: Recommended heave-data fields and units.

Field	Unit	Comment
`omega_rad_s`	rad/s	Use angular frequency, not cyclic frequency in Hz.
`A33_kg`	kg	Added mass in mode 3 due to unit heave acceleration.
`B33_Nsm`	N s/m	Radiation damping in heave.
`Fexc3_real_Npm`	N/m	Real part of excitation force per wave amplitude.
`Fexc3_imag_Npm`	N/m	Imaginary part under the selected harmonic convention.

7.3 Example heave-data import pattern

The safest workflow is to sort, validate, and interpolate the BEM data to the demo frequency vector. Avoid extrapolation when possible; extend the BEM frequency range instead.

```
% Inside calculateHeaveResponse(), after D.omega is defined:
tbl = readtable('heave_hydrodynamics.csv');

% Sort by angular frequency and preserve complex excitation phase.
[omegaData, idx] = sort(tbl.omega_rad_s);
A33Data = tbl.A33_kg(idx);
B33Data = tbl.B33_Nsm(idx);
FexcData = tbl.Fexc3_real_Npm(idx) ...
          + 1i*tbl.Fexc3_imag_Npm(idx);

% Prefer a BEM range that covers all D.omega values.
assert(min(D.omega) >= min(omegaData) && ...
       max(D.omega) <= max(omegaData), ...
       'BEM frequency range does not cover the demo range.');

D.addedMass = interp1(omegaData, A33Data, D.omega, 'pchip');
D.radiationDamping = interp1(omegaData, B33Data, D.omega, 'pchip');
D.excitationPerAmplitude = interp1(omegaData, FexcData, ...
                                   D.omega, 'pchip');
```

Listing 7.1: Illustrative import pattern for device-specific heave data.

Because `D.period` increases while `D.omega=2*pi./D.period` decreases, interpolation should be performed against a sorted frequency vector or against period consistently. Do not assume the source file uses the same ordering as the plots.

7.4 Minimum data required for the coupled demo

For heave-pitch coupling, provide the complete matrices and complex excitation vector:

$$\boldsymbol{A}(\omega_j) = \begin{bmatrix} A_{33} & A_{35} \\ A_{53} & A_{55} \end{bmatrix}, \qquad \boldsymbol{B}(\omega_j) = \begin{bmatrix} B_{33} & B_{35} \\ B_{53} & B_{55} \end{bmatrix}, \tag{7.2}$$

$$\frac{\hat{\boldsymbol{F}}_{\mathrm{exc}}}{a} = \begin{bmatrix} \hat{F}_{\mathrm{exc},3}/a \\ \hat{M}_{\mathrm{exc},5}/a \end{bmatrix}. \tag{7.3}$$

The body mass matrix, hydrostatic matrix, and rotation reference point must be consistent with the BEM definition of modes 3 and 5.

Table 7.2: Recommended coupled-data fields and units.

Field group	Unit	Comment
A_{33}	kg	Heave force per heave acceleration.
A_{35}, A_{53}	kg m	Translation-rotation added-mass coupling.
A_{55}	kg m^2	Pitch moment per pitch acceleration.
B_{33}	N s/m	Heave radiation damping.
B_{35}, B_{53}	N s	Translation-rotation damping coupling.
B_{55}	N m s/rad	Pitch radiation damping.
$F_{\mathrm{exc},3}/a$	N/m	Complex force transfer.
$M_{\mathrm{exc},5}/a$	N m/m	Complex moment transfer; often reported as N when normalized by metres, but label the normalization explicitly.

7.5 Example coupled-data insertion

The following pattern assumes the imported arrays have already been interpolated to `D.omega`.

```
numberOfFrequencies = numel(D.omega);
D.A = zeros(2,2,numberOfFrequencies);
D.B = zeros(2,2,numberOfFrequencies);
D.excitation = zeros(2,numberOfFrequencies);

for j = 1:numberOfFrequencies
    D.A(:,:,j) = [A33(j), A35(j); A53(j), A55(j)];
    D.B(:,:,j) = [B33(j), B35(j); B53(j), B55(j)];
    D.excitation(:,j) = [Fexc3_per_a(j); Mexc5_per_a(j)];
end

for j = 1:numberOfFrequencies
    Z = -D.omega(j)^2*(D.Mbody + D.A(:,:,j)) ...
        + 1i*D.omega(j)*(D.B(:,:,j) + D.Bexternal) ...
        + D.totalStiffness;
    D.H(:,j) = Z \ D.excitation(:,j);
end
```

Listing 7.2: Illustrative assembly of imported coupled hydrodynamic arrays.

7.6 Convention checklist before importing data

Do not skip this checklist

A numerically smooth curve can still be physically wrong if its convention is inconsistent with the response model.

Confirm all of the following:

1. **Frequency variable:** rad/s versus Hz, and ascending versus descending order.

2. **Wave normalization:** unit wave amplitude, unit wave height, nondimensional excitation, or another solver-specific scale.
3. **Harmonic convention:** $e^{+\mathrm{i}\omega t}$ versus $e^{-\mathrm{i}\omega t}$.
4. **Mode signs:** positive heave direction and positive pitch rotation.
5. **Reference point:** the point about which pitch moment and pitch mode are defined.
6. **Water depth and wave heading:** must match the intended case.
7. **Density and gravity:** dimensionalization must use the same values as the response model.
8. **Rotation units:** radians in the equations, even if plots display degrees.
9. **Matrix ordering:** verify that modes 3 and 5 have not been swapped.
10. **Complex excitation:** import both real and imaginary parts or magnitude and phase.

Capytaine documentation notes that different radiation-potential normalizations can differ by a factor involving $-\mathrm{i}\omega$ while yielding the same added mass and radiation damping under consistent processing. This is one reason to import final physical coefficients rather than mixing raw radiation potentials from different conventions.

7.7 Numerical and physical quality checks

For every frequency, check:

$$\boldsymbol{A} \approx \boldsymbol{A}^{\mathsf{T}}, \tag{7.4}$$

$$\boldsymbol{B} \approx \boldsymbol{B}^{\mathsf{T}}, \tag{7.5}$$

$$\lambda_{\min}(\boldsymbol{B}) \geq -\varepsilon, \tag{7.6}$$

$$\lambda_{\min}(\boldsymbol{C} + \boldsymbol{K}_{\text{extra}}) > 0 \tag{7.7}$$

for a justified numerical tolerance ε. Small asymmetry can arise from numerical discretization, but large asymmetry is a warning about mesh quality, mode definitions, or post-processing.

Also examine:

- mesh convergence of A, B, and excitation;
- expected low- and high-frequency trends;
- positivity of diagonal radiation damping;
- smoothness without spurious oscillations;
- phase continuity after deliberate unwrapping;
- hydrostatic coefficients from an independent calculation;
- mass and inertia against CAD or measurement;
- comparison with a limiting analytical solution, another BEM solver, or experiment; and
- sensitivity of RAOs to interpolation method and frequency-grid resolution.

7.8 Recommended validation ladder

1. **Geometry and equilibrium:** verify displacement, center of gravity, waterplane, draft, and reference point.
2. **Hydrostatics:** independently calculate C_{33} and check pitch restoring.
3. **Radiation sanity:** inspect diagonal signs, symmetry, matrix eigenvalues, and frequency limits.
4. **Diffraction sanity:** inspect complex force/moment magnitude and phase versus heading and frequency.
5. **Mesh convergence:** refine the wetted-surface mesh and quantify coefficient change.
6. **Cross-code or analytical comparison:** compare at least selected frequencies.
7. **Response comparison:** compare RAO peak, phase, and bandwidth.
8. **Experiment:** validate motion and loads at representative amplitudes; identify viscous/nonlinear departures.
9. **Time-domain handoff:** only after the frequency-domain model is trusted, construct radiation-memory or state-space models for irregular-wave simulations.

7.9 Relationship to WEC-Sim

The present demos solve a steady frequency-domain problem. WEC-Sim uses frequency-domain hydrodynamic coefficients in a time-domain wave-energy-converter model that can include rigid bodies, PTOs, moorings, viscous terms, constraints, and controls. A sensible classroom sequence is:

1. calculate or import $A(\omega)$, $B(\omega)$, and complex excitation;
2. inspect them in the present RAO demos;
3. verify the resulting frequency-domain response; and
4. transfer validated hydrodynamic data into the WEC-Sim/BEMIO workflow for time-domain studies.

Do not compare dimensional and nondimensional coefficients directly

WEC-Sim data structures and individual BEM solvers can store nondimensional coefficients or apply density, gravity, and frequency scaling. Always follow the specific tool documentation and verify a reconstructed physical coefficient in SI units before comparing it with the matrices used in this manual.

Chapter 8

Code Architecture and Customization

8.1 Design philosophy

Each demo is self-contained in one MATLAB function file. The top-level function creates the user interface and nested callbacks. Local functions at the end of the file define default parameters, calculate the model, render motion, provide prediction questions, run diagnostics, and execute self-tests.

This structure makes classroom distribution simple. For research batch processing, it may be preferable to move the calculation functions into separate files so they can be called directly without creating the GUI.

8.2 Function map: heave demo

Table 8.1: Local functions in `HMEC_heave_RAO_demo.m`.

Function	Responsibility
`HMEC_heave_RAO_demo`	Parses normal or `'selftest'` mode, creates the GUI, reads controls, plots outputs, and handles buttons.
`defaultHeaveParameters`	Defines constants, default geometry, damping, scales, period range, wave amplitude, and selected period.
`calculateHeaveResponse`	Constructs teaching curves, assembles impedance, calculates RAO and power, estimates balance period, and interpolates selected-period values.
`renderHeaveFrame`	Draws the wave, body, equivalent spring/dashpot, and selected harmonic state.
`drawSpring`, `drawDashpot`	Draw equivalent dynamic symbols.
`dispersionWavenumber`	Solves $\omega^2 = gk \tanh kh$.
`phaseDegrees`	Converts complex phase to degrees.
`heaveQuestionTitles`, `heaveQuestionText`	Define prediction prompts and explanations.
`heaveDiagnostics`	Runs fast model checks used by the GUI status label.
`runHeaveSelfTest`	Runs non-GUI trend and invariance checks.

8.3 Function map: coupled demo

Table 8.2: Local functions in `HMEC_heave_pitch_RAO_demo.m`.

Function	Responsibility
`HMEC_heave_pitch_RAO_demo`	Creates the coupled interface, reads controls, plots all tabs, and handles comparison and animation.
`defaultHeavePitchParameters`	Defines default body, matrix, damping, coupling, wave, and display parameters.
`calculateHeavePitchResponse`	Constructs $\boldsymbol{M}$, $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, excitation, impedance, RAOs, powers, and selected-period matrices.
`interpolateMatrix`	Interpolates each 2×2 entry at the selected period.
`scaleGeneralizedMatrix`	Transforms the pitch coordinate to $L\theta$ for comparable impedance units.
`renderHeavePitchFrame`	Draws simultaneous wave, heave, and pitch motion.
`pitchDisplayScale`	Converts rad/m to the chosen display units.
`heavePitchQuestionTitles`, `heavePitchQuestionText`	Define coupled prediction prompts and explanations.
`heavePitchDiagnostics`	Runs symmetry, definiteness, response, and period checks.
`runHeavePitchSelfTest`	Runs non-GUI coupled trend, transfer, and amplitude-invariance checks.

8.4 Changing default values

Edit only the relevant default-parameter function. For example:

```
function P = defaultHeaveParameters()
    ...
    P.ptoDamping_kNsPerM = 80;  % new default [kN s/m]
    ...
end
```

Listing 8.1: Example: changing the default heave PTO damping.

After changing a default, confirm that the corresponding spinner limits still contain the value and rerun the self-test.

8.5 Changing control ranges

The spinner limits are defined in the top-level GUI function. A control call follows the pattern

```
ptoDampingSpinner = makeSpinner(controls, rowNumber, ...
    'PTO damping [kN s/m]', [minimum maximum], step, ...
    defaults.ptoDamping_kNsPerM, @updateAll);
```

Listing 8.2: Spinner-definition pattern.

Large ranges can create near-singular response peaks, unrealistic motions, or difficult plot scaling. Expand limits only with a clear teaching or research purpose.

8.6 Adding a new prediction question

For the heave demo:

1. add a title string in `heaveQuestionTitles`;
2. add a matching branch in `heaveQuestionText`;
3. identify the control to change and the expected directional result; and
4. test the prompt, reveal button, and reset behavior.

Use the analogous coupled functions for the heave-pitch demo.

A strong question has four elements:

- one parameter change;
- a specific predicted output;
- a direct link to an equation term; and
- one caveat explaining why a real device may be more complicated.

8.7 Saving calculated data

The current code stores results inside callbacks. To export the latest case, add a button callback after `D = calculate...` has been evaluated. A simple pattern is:

```
out = table(D.period(:), D.omega(:), D.addedMass(:), ...
    D.radiationDamping(:), real(D.excitationPerAmplitude(:)), ...
    imag(D.excitationPerAmplitude(:)), abs(D.H(:)), ...
    angle(D.H(:))*180/pi, D.meanPtoPower(:), ...
    'VariableNames', {'T_s','omega_rad_s','A33_kg','B33_Nsm', ...
    'FexcReal_Npm','FexcImag_Npm','H3_mag','H3_phase_deg', ...
    'PtoPower_W'});
writetable(out,'heave_demo_case.csv');
```

Listing 8.3: Illustrative export of a heave case.

For the coupled case, export each matrix entry and both complex response components. Include the rotation reference point and sign conventions in the accompanying metadata.

8.8 Creating a non-GUI batch version

Because the calculation functions are local to the GUI files, they are not directly callable from another script. For repeated parameter sweeps:

1. copy `defaultHeaveParameters` and `calculateHeaveResponse` into separate function files;
2. preserve all units and parameter names;
3. write unit tests that compare the refactored output with the GUI self-test; and
4. let the GUI call the same shared calculation function to avoid two diverging implementations.

8.9 Adding another degree of freedom

The coupled model generalizes naturally:

$$\boldsymbol{Z}(\omega)\hat{\boldsymbol{q}} = \hat{\boldsymbol{F}}_{\mathrm{exc}} \tag{8.1}$$

for N degrees of freedom. Adding surge, for example, requires:

- expanding $\boldsymbol{M}$, $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, and $\boldsymbol{F}_{\mathrm{exc}}$;
- defining the correct units of every generalized entry;
- preserving mode signs and reference point;
- adding controls and plots without hiding matrix behavior; and
- extending symmetry, definiteness, and trend tests.

Do not simply append a diagonal scalar response if the geometry creates meaningful cross coupling.

8.10 Changing the harmonic convention

The demos use $e^{+\mathrm{i}\omega t}$, producing $+\mathrm{i}\omega\boldsymbol{B}$ in the impedance. If the code is changed to $e^{-\mathrm{i}\omega t}$, the damping sign and complex excitation convention must change consistently. Never change only the plot phase sign.

8.11 Coding practices to preserve

- use SI units internally;
- use MATLAB backslash for matrix solves;
- keep coefficient and excitation arrays on one common frequency vector;
- retain complex excitation values;
- avoid hidden global variables;
- keep the self-test independent of GUI creation;
- preserve explicit warning text that teaching curves are not device data; and
- update the manual and citation version when distributing a materially changed release.

Troubleshooting

9.1 Quick diagnostic sequence

1. Confirm all three `.m` files are in the Current Folder.
2. Run `which <function name> -all` to detect duplicates.
3. Run the relevant `'selftest'` command.
4. Reset the interface to default values.
5. Close and reopen the demo.
6. Restart MATLAB if graphics or cached-function behavior persists.

9.2 Symptom table

Table 9.1: Common symptoms, likely causes, and recommended actions.

Symptom	Likely cause	Action
"Undefined function" or launcher not found	File is not in Current Folder/path, or filename changed	Put all core files together; run `which ... -all`; confirm function and filename match exactly.
An older interface opens	Duplicate file earlier on MATLAB path	Use `which ... -all`; remove or rename stale copies; run `clear functions`.
Self-test passes but GUI fails	MATLAB release or desktop-graphics issue	Run on a recent desktop MATLAB release; capture the full error and line number; test a simple `uifigure`.
Error involving `UIAxes` and `grid`	An older or modified file may call `grid(ax,'on')` in a context MATLAB interprets incorrectly	Use the supplied version, which sets UIAxes grid properties directly; remove stale copies from the path.
Blank or incomplete plots	Extreme parameter combination, hidden window, graphics state, or update callback failure	Reset defaults; resize/maximize the window; rerun self-test; inspect the first error in the Command Window.
Very large RAO peak	Low damping near the inertia-restoring balance	This may be mathematically consistent but physically unrealistic; restore damping, note missing nonlinear limits, and avoid presenting it as a design result.
RAO does not change when wave amplitude changes	This is expected in linear theory	Compare the selected physical motion and PTO power instead.
Pitch remains when direct pitch moment is off	Coupling transfers heave forcing into pitch	Set coupling to 0% as a second test. Pitch should then vanish in this model.

Symptom	Likely cause	Action
A cross damping coefficient is negative	Cross terms can have either sign	Check full-matrix symmetry and eigenvalues; do not judge passivity from one entry alone.
Phase jumps from +180 to -180 degrees	Wrapped phase display	Interpret as a continuous phase crossing; unwrap only when a continuous phase curve is required.
Peak period differs from the displayed balance period	Excitation, damping, and coupling shift the actual RAO maximum	Use the balance as an interpretation aid, not an exact peak formula.
PTO power decreases when damping is increased	Higher damping has suppressed velocity enough to reduce $B\|\dot{q}\|^2/2$	Discuss the damping-power optimum and compare both power and RAO plots.
Pitch values look much larger after changing units	deg/m is $180/\pi$ times rad/m	Confirm the units label; internal dynamics are unchanged.
Imported BEM phase looks reversed	Harmonic convention or excitation conjugation mismatch	Check $e^{+\mathrm{i}\omega t}$ versus $e^{-\mathrm{i}\omega t}$, mode signs, and real/imaginary columns.
Imported matrices are not symmetric	Mesh error, inconsistent reference point, output mapping, or numerical tolerance	Inspect raw solver output, convergence, mode order, and dimensionalization before forcing symmetry.
Selected animation wavelength seems wrong	Water depth or frequency units are inconsistent	Verify period in seconds, $\omega = 2\pi/T$, depth in metres, and dispersion relation.

9.3 Interpreting a failed self-test

A self-test failure should not be bypassed. Identify which assertion failed:

- **finite response failure**: inspect zero/negative stiffness, NaNs, or a singular matrix;
- **sign failure**: inspect coefficient formulas and unit conversions;
- **trend failure**: confirm only the intended parameter changed;
- **symmetry/PSD failure**: inspect cross-correlation magnitude and matrix assembly;
- **amplitude-invariance failure**: a nonlinear or amplitude-dependent term may have been introduced; or
- **coupling-transfer failure**: inspect off-diagonal matrix terms and direct excitation switch logic.

9.4 Reporting a reproducible issue

Record:

- MATLAB release and operating system;
- exact filename and file date;
- the output of `which ... -all`;
- the full error message and line number;

- whether the self-test passes;
- the input values that reproduce the problem; and
- a screenshot of the relevant tab.

Instructor Quick Reference

10.1 Before students arrive

Five-minute readiness check

1. Run both self-tests.
2. Open both demos.
3. Confirm baseline comparison and animation.
4. Reset defaults.
5. Display the governing equation on the lecture slide before opening the GUI.

10.2 Heave question-and-answer guide

Table 10.1: Fast instructor guide for 1-DOF prediction questions.

Change	Expected first-order trend	Equation-based explanation
Increase body mass	Longer-period response region	Larger $m + A_{33}$ requires lower ω to balance fixed stiffness.
Increase added mass scale	Longer-period response region	Hydrodynamic effective inertia increases.
Increase radius	Usually shorter in this teaching model	$C_{33} = \rho g \pi r^2$ grows strongly; real geometry also changes all hydrodynamic curves.
Increase PTO damping	Lower, broader RAO peak	Imaginary impedance grows; power may have an optimum.
Increase extra stiffness	Shorter-period response region	Larger restoring requires higher ω for balance.
Double wave amplitude	RAO unchanged; motion $\times 2$; power $\times 4$	Linear excitation and response scale together; damper power scales with velocity squared.

10.3 Coupled question-and-answer guide

Table 10.2: Fast instructor guide for coupled prediction questions.

Change	Expected first-order trend	Equation-based explanation
Set coupling to 0%	Equations separate	$A_{35} = B_{35} = C_{35} = 0$.
Turn off direct pitch moment	Pitch may remain	Heave forcing can transfer through off-diagonal impedance.
Turn off direct moment and coupling	Pitch becomes zero	Pitch equation is then unforced and independent.
Increase I_{yy}	Pitch response shifts longer	Effective rotational inertia increases.
Increase pitch stiffness	Pitch response shifts shorter	Rotational restoring increases.
Increase pitch PTO damping	Pitch peak decreases/broadens	Rotational damping term grows; power remains an optimization tradeoff.
Increase heave stiffness	Heave changes directly; pitch can change indirectly	The full matrix solution redistributes response.
Switch deg/m to rad/m	Display scale changes only	Internal solution remains in radians.

10.4 One-slide verbal summary

Suggested closing script

"The waves provide a complex force and moment. The body responds according to a complex impedance made from rigid-body inertia, added mass, radiation and PTO damping, and hydrostatic or mechanical restoring. In heave, this is one denominator. In heave and pitch, it is a matrix. Changing mass or stiffness mainly shifts the balance; changing damping reshapes the peak; coupling allows one forcing component to produce another motion. The RAO describes response per unit wave amplitude, while actual motion and ideal mechanical power also depend on wave amplitude."

10.5 Suggested student exercises

1. Estimate the heave balance period by evaluating Equation (4.12) at a selected coefficient value and compare with the RAO peak.
2. Find a PTO damping that maximizes ideal mechanical power at one chosen period, then explain why it may not maximize energy over a spectrum.
3. Construct a two-case table showing how mass and stiffness produce opposite period shifts.
4. In the coupled demo, remove direct pitch excitation and quantify the remaining pitch caused by coupling.
5. Convert the pitch RAO from rad/m to deg/m manually and verify the interface.
6. Export one synthetic coefficient set and reproduce the RAO in a separate MATLAB script.
7. Replace the heave teaching curves with a supplied BEM dataset and complete the convention checklist in Chapter 7.

10.6 Recommended message before students use the controls

Say this explicitly

“The curves are physically constrained teaching data, not a prediction for a particular WEC. Today we are learning how coefficients combine to create motion. In a device analysis, we would replace these curves with converged hydrodynamic data for the actual geometry and validate the response.”

Numerical Validation Summary

The equations were independently reproduced in Python/NumPy using the same parameter values and formulas. The checks below passed. They verify algebraic implementation and expected trends; they do not validate a real device.

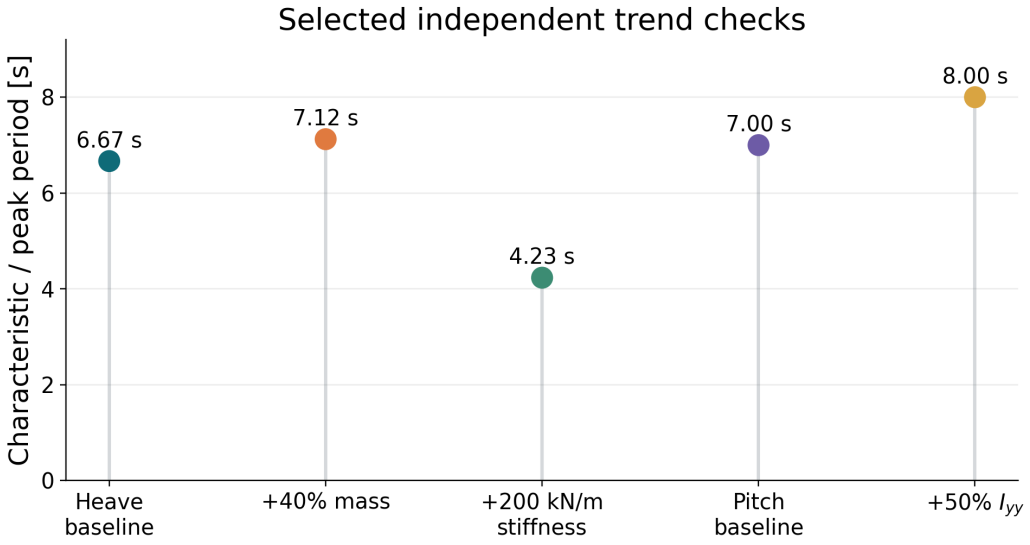


Figure A.1: Selected independently reproduced period-shift checks.

A.1 1-DOF heave checks

Table A.1: Independent 1-DOF numerical checks.

Check	Result
Default mass	115.925 tonne
$C_{33} = \rho g A_{wp}$	284.305 kN/m
A_{33} range	74.961–96.654 tonne
Maximum B_{33}	119.818 kN s/m
$\lvert F_{\mathrm{exc},3}\rvert/a$ range	150.620–326.068 kN/m
Default RAO peak	T = 6.667 s, $\lvert H_3\rvert$ = 1.528973
Increase mass by 40%	Peak moves to 7.124 s: pass
Add 200 kN/m stiffness	Peak moves to 4.234 s: pass
High PTO damping	Peak drops to 1.126463: pass
Double wave amplitude	Maximum RAO change 0.000×10^{0}: pass

A.2 Coupled heave-pitch checks

Table A.2: Independent coupled numerical checks.

Check	Result
Default mass and pitch inertia	420 tonne; 13.00×10^6 kg m^2
Reference draft	5.122 m
Restoring entries	C_{33} = 804.420 kN/m; C_{55} = 18.000 MN m/rad; C_{35} = 380.520 kN
A_{33} range	230.852–329.408 tonne
A_{55} range	$7.559 - -9.863 \times 10^6$ kg m^2
Maximum radiation damping	B_{33} = 319.689 kN s/m; B_{55} = 5.199 MN m s/rad
Default heave peak	T = 6.698 s, $\lvert H_3 \rvert$ = 1.949580 m/m
Default pitch peak	T = 6.349 s, $\lvert H_5 \rvert$ = 0.170528 rad/m = 9.770510 deg/m
Minimum determinant of $\boldsymbol{A}$	$1.741724 \times 10^{12} > 0$: pass
Minimum determinant of $\boldsymbol{B}$	$3.511865 \times 10^8 > 0$: pass
Eigenvalues of $\boldsymbol{C} + \boldsymbol{K}$	7.960036×10^5, 1.800842×10^7: pass
Coupling 0 and direct moment off	Maximum $\lvert H_5 \rvert$ = 0: pass
Coupling on and direct moment off	Maximum $\lvert H_5 \rvert$ = 0.072559 rad/m: transfer pass
Increase I_{yy} by 50%	Pitch balance 7.000 s to 8.000 s: pass
High pitch damping	Peak 0.105767 < 0.170528 rad/m: pass
Add 600 kN/m heave stiffness	Heave balance 5.977 s to 4.349 s: pass
Double wave amplitude	Maximum change in either RAO 0.000×10^0: pass

A.3 Static source-file checks

The three MATLAB files were also checked for balanced parentheses, brackets, and braces and for ASCII-only source text. All checks passed. The remaining environment-specific check is launching the `uifigure` interfaces in the target MATLAB release.

Symbol and Unit Reference

Table B.1: Symbols used in the demos and manual.

Symbol	Typical unit	Meaning
$a, \hat{a}$	m	Incident-wave amplitude and complex amplitude.
A_{ij}	generalized mass units	Added-mass coefficient: generalized force in mode i due to acceleration in radiating mode j.
A_{wp}	m^2	Waterplane area.
B_{ij}	generalized damping units	Radiation-damping coefficient.
B_{PTO}	N s/m or N m s/rad	Linear PTO damping.
B_{visc}	N s/m or N m s/rad	Equivalent linear viscous damping.
C_{ij}	generalized stiffness units	Hydrostatic restoring coefficient.
d	m	Reference draft.
$F_{\mathrm{exc},3}$	N	Heave excitation force.
g	m/s^2	Gravitational acceleration.
H_3	m/m	Heave RAO.
H_5	rad/m or deg/m	Pitch RAO.
h	m	Water depth.
I_{yy}	kg m^2	Rigid-body pitch inertia about the chosen reference point.
k	rad/m	Wave number.
K_{extra}	generalized stiffness units	Additional linear PTO, mooring, or tuning stiffness.
L	m	Reference body length.
m	kg	Rigid-body mass.
$M_{\mathrm{exc},5}$	N m	Pitch excitation moment.
P_{PTO}	W	Mean ideal mechanical PTO power.
$q, \boldsymbol{q}$	m and rad	Generalized coordinate or coordinate vector.
r	m	Waterplane radius in the heave demo.
T	s	Wave period.
$Z, \boldsymbol{Z}$	dynamic-stiffness units	Scalar or matrix complex impedance.
z	m	Heave displacement, positive upward.
θ	rad	Pitch rotation.
η	m	Incident wave elevation.
ρ	kg/m^3	Water density.
ω	rad/s	Angular wave frequency, $2\pi/T$.

Package File Manifest

A complete manual package should contain:

- this compiled PDF;
- the editable `HMEC_RAO_Demos_User_Manual.tex`;
- `HMEC_RAO_Demos_References.bib`;
- the `assets` folder used by the manual;
- the `matlab` folder containing the three core demos;
- the `source_material` folder containing the README, validation report, and presentation recommendations; and
- `README_COMPILE.txt`.

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End of manual - Version 1.0 - August 2026